# AIR POLLUTION AND CONTROL

## Subject Code : 23A01504b

## UNIT - I : AIR POLLUTION

**Syllabus:** Definition - Sources & Classification of Air Pollutants - Effects on Humans, Plants and Materials - Global Effects - Air Quality and NAAQS - National Clean Air Programme - Sampling of Pollutants in Ambient Air - Stack Sampling.

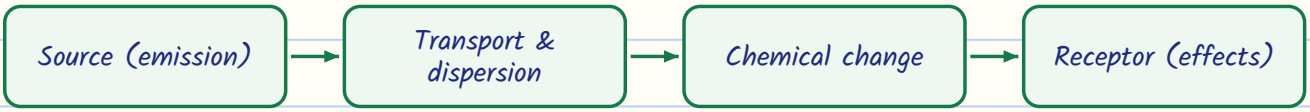
## 1. Air Pollution - Definition

Air is a mixture of gases: $N_2$ (78%), $O_2$ (21%), Ar (0.93%), $CO_2$ (0.03-0.04%) and traces of other gases plus water vapour. When the natural composition of air is disturbed, we call it pollution.

**DEFINITION: Air pollution is the presence in the outdoor atmosphere of one or more contaminants (dust, fumes, gas, mist, odour, smoke, vapour) in such quantity and of such duration that they are or may tend to be injurious to human, plant or animal life, or to property, or which unreasonably interfere with the comfortable enjoyment of life.**

**Air Act, 1981 (India):** 'Air pollutant' means any solid, liquid or gaseous substance (including noise) present in the atmosphere in such concentration as may be or tend to be injurious to human beings, other living creatures, plants, property or environment.

- Air pollution = Source + Atmosphere (transport, dilution, chemical change) + Receptor
- Pollutants are measured in $ug/m^3$, $mg/m^3$ (mass/volume) or ppm, ppb (volume/volume)
- Conversion: $ug/m^3$ = (ppm x mol. wt. x 1000) / 24.45 at 25 deg C and 1 atm




## 2. Sources of Air Pollutants

### (a) Natural sources

- Volcanic eruptions - ash, $SO_2$, $H_2S$, $CO_2$
- Forest fires - smoke, CO, $CO_2$, particulates
- Dust storms and wind-blown soil; sea spray (salt particles)
- Pollen, spores, bacteria; decay of organic matter (methane, $H_2S$)
- Lightning - forms NO from $N_2$ and $O_2$; cosmic dust, meteors

### (b) Anthropogenic (man-made) sources

- **Combustion of fuels** - thermal power plants, industries, domestic cooking and heating
- **Transportation** - cars, trucks, buses, ships, aircraft, railways (CO, $NO_x$, HC, Pb, PM)
- **Industrial processes** - cement, steel, fertiliser, refineries, chemical, paper, smelters
- **Agriculture** - pesticide spray, burning of crop residue, $NH_3$ from fertilisers, $CH_4$ from paddy and cattle
- **Solid waste** - open burning, incinerators, landfill gas
- **Miscellaneous** - construction, mining, quarrying, wars and nuclear explosions

### (c) Classification of sources by geometry

| Type | Description | Examples |
|---|---|---|
| Point source | Single fixed, identifiable location | Chimney, stack of power plant |
| Line source | Emission along a line / road | Highway, railway track |
| Area source | Many small sources over a region | City, domestic burning, landfill |
| Mobile source | Moves from place to place | Vehicles, ships, aircraft |
| Stationary source | Fixed in position | Factories, boilers, furnaces |

## 3. Classification of Air Pollutants

### (i) Based on origin

- **Primary pollutants** - emitted directly from the source. e.g. $SO_2$, NO, CO, soot, fly ash, Pb, HC
- **Secondary pollutants** - formed in the atmosphere by reaction of primary pollutants. e.g. ozone ($O_3$), PAN (peroxyacetyl nitrate), sulphuric acid mist, nitric acid, aldehydes

**Remember: Smog (photochemical) = $NO_x$ + HC + sunlight -> $O_3$ + PAN + aldehydes (all secondary)**

### (ii) Based on physical state

- **Gaseous pollutants** - $SO_2$, $NO_x$, CO, $CO_2$, $H_2S$, $NH_3$, HF, $O_3$, VOCs
- **Particulate pollutants** - solids or liquid droplets: dust, smoke, fumes, mist, fog, fly ash, aerosols

## (iii) Particulate matter - terms

| Term | Meaning |
|---|---|
| Dust | Solid particles by crushing/grinding, size > 1 um (settle by gravity) |
| Smoke | Fine solid/liquid particles from incomplete combustion, < 1 um |
| Fume | Solid particles by condensation of vapours (metal fumes), 0.03-0.3 um |
| Mist / Fog | Liquid droplets by condensation or atomisation; fog is visible mist |
| Fly ash | Non-combustible residue carried by flue gas of coal furnaces |
| Aerosol | Any suspension of fine solid/liquid particles in gas |
| PM10 / PM2.5 | Particles with aerodynamic diameter < 10 um / < 2.5 um |

## (iv) Based on chemical nature

- Inorganic: oxides of S, N, C; halogens; metals (Pb, Hg, Cd)
- Organic: hydrocarbons, aldehydes, ketones, PAHs, benzene, VOCs
- Radioactive: radon, iodine-131, strontium-90

## (v) Criteria pollutants

Pollutants for which standards are fixed on the basis of health criteria: **$SO_2$, $NO_2$, CO, $O_3$, Pb, PM10 and PM2.5** (US-EPA list of six; India's NAAQS covers 12).

# 4. Major Pollutants - Sources and Effects

| Pollutant | Main sources | Main effects |
|---|---|---|
| CO | Vehicle exhaust, incomplete combustion | Binds Hb (carboxyhaemoglobin), headache, death |
| $SO_2$ | Coal / oil burning, smelters | Bronchitis, acid rain, leaf damage |
| $NO_x$ | Vehicles, power plants (high temp.) | Lung irritation, smog, acid rain |
| Hydrocarbons | Vehicles, solvents, refineries | Smog, cancer (benzene, PAH) |
| Lead | Leaded petrol, battery, paint | Brain damage in children, anaemia |
| $O_3$ | Secondary (photochemical) | Eye/lung irritation, crop damage |
| Particulates | Industry, dust, vehicles, burning | Asthma, silicosis, visibility loss |

# 5. Effects of Air Pollution

## 5.1 Effects on human health

Pollutants enter the body mainly by **inhalation**. The nose filters particles > 10 um; particles of 2.5-10 um reach the bronchi; PM2.5 and smaller reach the alveoli and may even enter the blood.

- **Carbon monoxide:** affinity for haemoglobin is about 200-250 times that of oxygen. COHb reduces $O_2$ carrying capacity -> headache, dizziness, coma, death at ~ 1000 ppm.
- **$SO_2$:** irritates the upper respiratory tract; chronic bronchitis, asthma, emphysema. It combines with PM and the effect is synergistic.
- **$NO_2$:** deep lung irritant; pulmonary oedema; reduces resistance to infection; forms nitrosamines.
- **Particulates:** silicosis (silica), asbestosis, black lung (coal), byssinosis (cotton), lung cancer, heart disease.

- **Lead:** damages nervous system, kidney; lowers IQ in children. **Benzene:** leukaemia.
- **Ozone / PAN:** eye irritation, cough, chest pain, reduced lung function.
- **Fluorides:** fluorosis of bones and teeth. **Mercury/Cadmium:** Minamata and Itai-itai disease.

**Famous episodes: Meuse Valley (1930), Donora (1948), London smog (1952, ~4000 deaths), Los Angeles smog (1940s), Bhopal gas tragedy (Dec 1984, methyl isocyanate - MIC).**

## 5.2 Effects on plants

- **$SO_2$** - chlorosis (yellowing) and necrosis between veins; bleached spots. Sensitive: alfalfa, barley, cotton
- **Ozone** - flecking and stippling on upper leaf surface; tobacco, spinach, tomato are sensitive
- **PAN** - glazing / silvering on lower leaf surface; young leaves most affected
- **Fluorides** - tip burn, leaf margin necrosis; very toxic to plants even at ppb level (gladiolus, apricot)
- **Ethylene** - leaf fall (abscission), withered flowers; $NO_2$ - growth suppression
- **Particulates** - dust layer blocks stomata and sunlight, reducing photosynthesis (cement dust)
- Acid rain damages leaves and leaches nutrients from soil; forest die-back

## 5.3 Effects on materials

- **Metals:** $SO_2$ + moisture -> $H_2SO_4$; corrosion of iron, zinc, copper, aluminium
- **Building stones:** limestone, marble: $CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + CO_2$ (the Taj Mahal 'stone cancer')

- **Paints:** darkening and blistering by $H_2S$ (lead paint turns black as PbS)
- **Textiles, paper, leather:** weakened and discoloured by acid gases
- **Rubber:** cracks due to ozone (tyres, hoses); dyes fade due to $NO_x$
- Soiling of buildings by soot and dust: heavy cleaning cost

### 5.4 Effects on animals and atmosphere

- Fluorosis in cattle, arsenic and lead poisoning through grazing on contaminated grass
- Reduction in visibility (haze), less sunlight, change in local climate (urban heat island)

## 6. Global Effects of Air Pollution

### 6.1 Greenhouse effect and global warming

Short-wave solar radiation reaches earth; the earth re-radiates long-wave infrared, which is absorbed by **greenhouse gases (GHGs)** - $CO_2$, $CH_4$, $N_2O$, CFCs, $O_3$, water vapour - and radiated back. Natural greenhouse effect keeps earth at +15 deg C instead of -18 deg C. Enhanced GHG levels cause **global warming**.

| GHG | Main source | Relative GWP (100 yr) |
|---|---|---|
| $CO_2$ | Fossil fuel burning, deforestation | 1 |
| $CH_4$ | Paddy, cattle, landfills, coal mines | about 28 |
| $N_2O$ | Fertilisers, biomass burning | about 265 |
| CFC-12 / HFCs | Refrigerants, aerosols | thousands (about 10,000) |
| $SF_6$ | Electrical industry | about 23,500 |

**Consequences:** sea-level rise, melting of glaciers, change in rainfall, droughts, floods, spread of diseases, loss of biodiversity, crop yield changes.

**Control:** Kyoto Protocol (1997), Paris Agreement (2015) - hold warming well below 2 deg C, preferably 1.5 deg C; renewable energy, afforestation, energy efficiency.

## 6.2 Ozone layer depletion

Stratospheric ozone (15-35 km) absorbs UV-B (280-315 nm). **Ozone depleting substances (ODS):** CFCs, halons, $CCl_4$, methyl chloroform, HCFCs.

UV splits CFC in the stratosphere releasing Cl atoms which act as a catalyst:

- $CF_2Cl_2 + UV \rightarrow CF_2Cl + Cl$
- $Cl + O_3 \rightarrow ClO + O_2$
- $ClO + O \rightarrow Cl + O_2$ (Cl is regenerated; one Cl destroys about 100,000 $O_3$ molecules)

**Ozone hole** over Antarctica appears in September-October due to polar stratospheric clouds. **Effects:** skin cancer, cataract, weak immunity, damage to phytoplankton and crops. **Dobson Unit (DU)** measures ozone column; the 'hole' is < 220 DU.

**Montreal Protocol (1987) phases out ODS; Kigali Amendment (2016) phases down HFCs.**

## 6.3 Acid rain

Rain with **pH < 5.6** (normal rain is slightly acidic due to dissolved $CO_2$). $SO_2$ and $NO_x$ are oxidised and dissolve in cloud water:

- $SO_2 + OH \rightarrow HOSO_2$; $SO_2 + 1/2\ O_2 + H_2O \rightarrow H_2SO_4$
- $NO_2 + OH \rightarrow HNO_3$
- **Effects:** acidification of lakes (fish die), forest decline, soil nutrient loss (Ca, Mg), corrosion of buildings and statues, human health

- **Control:** flue gas desulphurisation, low-sulphur fuels, catalytic converters, liming of lakes

### 6.4 Photochemical smog and other global issues

- Photochemical smog (Los Angeles type): $NO_2$ + sunlight -> NO + O; O + $O_2$ -> $O_3$; HC + $O_3$ -> PAN, aldehydes
- Classical (London) smog: smoke + fog + $SO_2$; cold, humid, reducing conditions
- Atmospheric brown cloud (ABC), black carbon on Himalayan snow -> faster melting
- Transboundary pollution, haze and loss of visibility

| Feature | London smog | Photochemical (LA) smog |
|---|---|---|
| Season / time | Winter, early morning | Summer, noon |
| Temperature | Low (-1 to 4 deg C) | High (24-32 deg C) |
| Main pollutants | Smoke, $SO_2$, fog | $NO_x$, HC, $O_3$, PAN |
| Nature | Reducing | Oxidising |
| Effect | Bronchitis, deaths | Eye irritation, crop damage |

## 7. Air Quality and NAAQS

**Ambient air quality** is judged against standards. **National Ambient Air Quality Standards (NAAQS)** are limits of pollutant concentration in outdoor air, set by the **CPCB** under Air Act 1981 (first in 1994, revised 2009, notified Nov 2009). Standards give two categories of area: (i) industrial, residential, rural and other areas and (ii) ecologically sensitive areas (notified by Central Government).

| Pollutant | Time-weighted avg. | Industrial/Residential | Eco-sensitive |
|---|---|---|---|
| $SO_2$ | Annual / 24 hr | 50 / 80 ug/$m^3$ | 20 / 80 |
| $NO_2$ | Annual / 24 hr | 40 / 80 ug/$m^3$ | 30 / 80 |

| Pollutant | Time-weighted avg. | Industrial/Residential | Eco-sensitive |
|---|---|---|---|
| PM10 | Annual / 24 hr | 60 / 100 $ug/m^3$ | 60 / 100 |
| PM2.5 | Annual / 24 hr | 40 / 60 $ug/m^3$ | 40 / 60 |
| Ozone | 8 hr / 1 hr | 100 / 180 $ug/m^3$ | 100 / 180 |
| Lead | Annual / 24 hr | 0.5 / 1.0 $ug/m^3$ | 0.5 / 1.0 |
| CO | 8 hr / 1 hr | 2 / 4 $mg/m^3$ | 2 / 4 |
| $NH_3$ | Annual / 24 hr | 100 / 400 $ug/m^3$ | 100 / 400 |

Others: Benzene 5 $ug/m^3$ (annual), Benzo(a)pyrene 1 $ng/m^3$, Arsenic 6 $ng/m^3$, Nickel 20 $ng/m^3$ (annual) - total 12 parameters.

### Air Quality Index (AQI) - India, launched Oct 2014

- Converts concentration of 8 pollutants (PM10, PM2.5, $NO_2$, $SO_2$, CO, $O_3$, $NH_3$, Pb) into one number 0-500
- Good 0-50 | Satisfactory 51-100 | Moderate 101-200 | Poor 201-300 | Very Poor 301-400 | Severe 401-500
- Sub-index calculated for each pollutant; **AQI = maximum sub-index** (at least 3 pollutants, one of them PM10 or PM2.5)

**Air Quality Management:** monitoring -> emission inventory -> standards -> modelling -> control strategy -> enforcement. WHO global guidelines (2021): PM2.5 annual 5 $ug/m^3$, 24 hr 15 $ug/m^3$.

## 8. National Clean Air Programme (NCAP)

Launched by the **MoEFCC in January 2019** - the first national, time-bound, city-specific plan to curb air pollution in India.

- **Goal:** 20-30% reduction in PM10 and PM2.5 concentration by 2024 (base year 2017); later revised to **40% reduction by 2025-26**
- **Target cities:** 131 non-attainment cities (cities that did not meet NAAQS for 5 years, 2011-2015) - includes cities in 24 states/UTs

- Each city prepares a **City Action Plan** - road dust, vehicular, industrial, biomass, construction, waste burning
- Plan is implemented through **city / state / central level committees**; PRANA portal for tracking
- Funding: Finance Commission grants for million-plus cities; National Clean Air Fund concept
- Strengthening of **monitoring network** (CAAQMS, manual stations) and source apportionment studies
- Public awareness, capacity building, technology assessment, emission inventory, health impact studies
- Complementary steps: BS-VI fuels (April 2020), GRAP in Delhi-NCR, Ujjwala LPG scheme, Pollution Under Control (PUC) certificates, EV promotion (FAME), ban on stubble burning, SAFAR forecasting

## 9. Sampling of Pollutants in Ambient Air

**Objectives:** to find the quality of ambient air, to compare with NAAQS, to identify trends and sources, to assess health effects and the effectiveness of control and to plan siting of industries.

### Principles of sampling

- Samples must be **representative** - collected at 1.5-10 m above ground (breathing zone), away from local obstructions (2 m from walls, trees)
- Number and location of stations depend on population, topography, wind rose, industrial areas, pollutant type
- Sampling duration: 24 hours (4-hourly or 8-hourly for gases), minimum 104 measurements per year (twice a week)
- **Grab sampling** gives instantaneous values; **continuous sampling** gives averages and trends

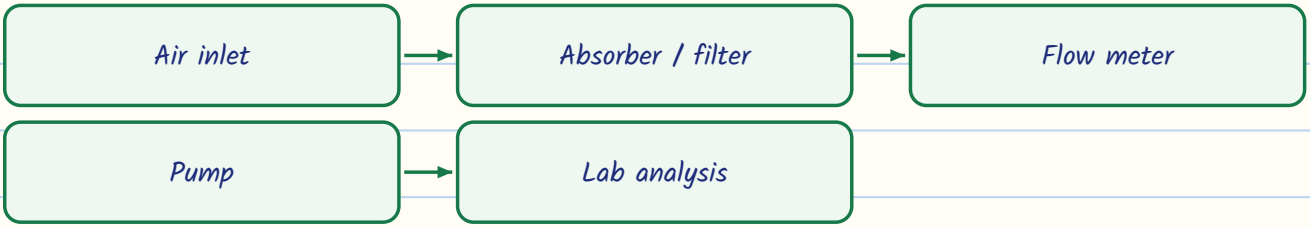
## (a) Particulate sampling - High Volume Sampler (HVS)

- Air drawn by a blower at **1.1-1.7 $m^3$/min** through a pre-weighed **glass-fibre filter** (20 x 25 cm) for 24 hr
- Mass concentration = $(W_2 - W_1) / V$ where V = flow rate x time (corrected to standard conditions)
- Collects SPM up to 100 um. For PM10 a **cyclone / impactor head** is fitted (Respirable Dust Sampler, RDS, flow 1.0-1.2 $m^3$/min)
- Filter weighed after conditioning in desiccator; also analysed for metals, sulphate, nitrate, BaP
- Others: Beta attenuation monitor (BAM), TEOM (oscillating microbalance), light scattering - **continuous**

## (b) Gaseous sampling

Gas is collected by bubbling through an absorbing solution in an **impinger / bubbler**, by adsorption on a solid (charcoal, silica gel), by condensation (cold trap) or **grab** in bags/bottles.

| Pollutant | Method | Absorbing reagent / principle |
|---|---|---|
| $SO_2$ | West-Gaeke (pararosaniline) | Sodium tetrachloromercurate; red-violet colour at 560 nm |
| $NO_2$ | Jacob-Hochheiser (modified) | Sodium hydroxide + sodium arsenite; azo dye, 540 nm |
| CO | Non-dispersive IR (NDIR) | IR absorption at 4.6 um |
| $O_3$ | Chemiluminescence / UV photometry | Reaction with ethylene or UV absorption 254 nm |
| $NH_3$ | Indophenol blue method | Dilute sulphuric acid, 630 nm |
| Pb | AAS / ICP | Extraction of filter paper in nitric acid |
| Hydrocarbons | Flame ionisation detector, GC | Carbon ions in hydrogen flame |




**Automatic continuous analysers (CAAQMS):** UV fluorescence ($SO_2$), chemiluminescence ($NO_x$), NDIR (CO), UV photometer ($O_3$) - data logged and sent online to CPCB.

## 10. Stack Sampling

Sampling of **flue gas inside a chimney/stack/duct** to measure the emission of particulate and gaseous pollutants from a source and to check compliance with emission standards (e.g. 150 mg/Nm$^3$ for particulates, as per CPCB).

### Why / when

- To check emission standards; to determine efficiency of control equipment; for design of control device; to find mass emission rate (kg/h)

### Sampling location (ports)

- In a straight vertical duct, **8 stack diameters downstream** and **2 diameters upstream** of any bend, fan, damper or obstruction
- Sampling ports (10 cm dia) at 90 deg apart; a sampling platform with safe access is needed
- **Traverse points:** the cross section is divided into equal areas (centre of each); small stacks need fewer points (minimum 8 for circular ducts up to 3 m dia)
- Velocity measured at every traverse point because velocity varies across the section

## Isokinetic sampling

**Isokinetic: the velocity of gas entering the sampling nozzle = velocity of gas in the stack at that point ($V_n = V_s$)**

- If $V_n > V_s$ (over-sampling): extra gas is drawn in, but heavy particles cannot turn and are missed -> **coarse particle conc. reads LOW**
- If $V_n < V_s$ (under-sampling): gas streams deflect around the nozzle but heavy particles go straight in by inertia -> **coarse particle conc. reads HIGH**
- Fine particles (< 5 um) follow gas streamlines, so error is small; isokinetic sampling gives a representative dust load

## Stack sampling train (EPA method 5 type)




1. **Nozzle & probe:** stainless steel/ glass-lined, nozzle sharp-edged, faced into the gas flow
2. **Pitot tube (S-type) + manometer:** measures velocity pressure
3. **Thermocouple:** measures stack gas temperature
4. **Filter holder (heated box):** collects particulate (at about 120 deg C to avoid condensation)
5. **Impingers:** collect moisture and gaseous pollutants ($SO_2$, $NO_x$) in absorbing solutions; silica gel dries the gas
6. **Pump, dry gas meter and orifice meter:** control and record sample volume; flow adjusted to remain isokinetic

### Calculations

- Stack gas velocity: $V_s = C_p \times \text{sqrt}( 2 \times g \times dP / \text{rho} )$ [$C_p$ = pitot coefficient (about 0.84 for S-type, 1.0 for standard)]
- Flow rate: $Q = V_s \times A$ ($m^3/s$); convert to normal conditions (0 deg C, 1 atm - $Nm^3$)
- Concentration C = mass of particulate on filter / volume of gas sampled (corrected to $Nm^3$, dry, 12% $CO_2$ or 7% $O_2$ reference)
- Emission rate (kg/h) = C x Q

**Gaseous stack sampling** is done by extractive probes with analysers (Orsat apparatus for $CO_2$, $O_2$, CO; flue gas analyser using electrochemical cells; Draeger tubes) or by in-situ / CEMS continuous emission monitoring.

**Ambient sampling -> measures air quality (low conc., ug/$m^3$). Stack sampling -> measures emissions (high conc., mg/$Nm^3$), needs isokinetic conditions.**

## II. Quick Revision / Important Questions

1. Define air pollution. Classify air pollutants with examples.
2. Explain the effects of air pollution on humans, plants and materials.
3. Write briefly on greenhouse effect, ozone depletion and acid rain.
4. What is NAAQS? List the standards for $SO_2$, $NO_2$, PM10 and PM2.5.
5. Describe the National Clean Air Programme (NCAP) and its objectives.
6. With a neat sketch describe the high-volume sampler.
7. What is isokinetic sampling? Explain a stack sampling train.

# AIR POLLUTION AND CONTROL

## Subject Code : 23A01504b

## UNIT - II : METEOROLOGY AND AIR POLLUTION

**Syllabus:** Factors influencing air pollution, Wind rose, Mixing depths, Lapse rates and dispersion - Atmospheric stability, Plume rise and dispersion, Prediction of air quality, Box model - Gaussian model - Dispersion coefficient - Application of tall chimney for pollutant dispersion.

## 1. Introduction

Once pollutants are emitted they are carried, diluted and transformed in the atmosphere. **Meteorology** (study of atmospheric processes) decides how far and how fast the pollutants are dispersed. The same emission can cause serious pollution on one day and negligible pollution on the next day, only because of the weather.

- Atmosphere layers: Troposphere (0-12 km, weather, pollution) | Stratosphere (12-50 km, ozone) | Mesosphere | Thermosphere
- Most air pollution problems occur in the lowest part of the troposphere - the **planetary boundary layer** (up to 1-2 km)
- Dispersion = transport (advection by wind) + dilution (turbulent diffusion)

## 2. Factors Influencing Air Pollution

### 2.1 Meteorological factors

- **Wind speed:** concentration is **inversely proportional to wind speed** ($C \sim 1/u$). High speed -> more dilution

- **Wind direction:** decides which area is polluted (downwind); persistent direction gives high dose at a place
- **Temperature:** affects stability, rate of chemical reactions, plume buoyancy; temperature inversion traps pollutants
- **Solar radiation:** drives convection and mixing; also drives photochemical reactions ($O_3$, PAN)
- **Humidity and fog:** $SO_2$ converts to sulphuric acid mist; fog and low cloud increase pollutant retention
- **Precipitation:** washout (below-cloud scavenging) and rainout (in-cloud) clean the air; acid rain results
- **Atmospheric stability and turbulence:** most important - decide vertical mixing

## 2.2 Topographical factors

- **Valleys:** cold air drains down at night and pollutants collect in the valley (inversion trap)
- **Hills / buildings:** cause downwash, eddies, wake effects; plume impinges on hill slope
- **Land and sea breezes:** sea breeze by day, land breeze by night - can recirculate pollutants (coastal)
- **Mountain-valley winds:** anabatic (up-slope, day) and katabatic (down-slope, night)
- **Urban heat island:** warmer city centre, rising air, inflow of air near ground (dust dome)
- **Vegetation, water bodies:** change surface roughness and heating

## 2.3 Source and receptor factors

- Emission rate, stack height, exit velocity, exit temperature, location of the source
- Nature of pollutant: particle size, density, chemical reactivity, settling velocity
- Distance and elevation of the receptor from the source, population density

**Pollution potential = Emission strength x Meteorological conditions x Topography**

# 3. Wind and Wind Rose

Wind is horizontal movement of air due to pressure gradient. **Wind speed** is measured by cup **anemometer**, **direction** by wind vane (direction FROM which the wind blows). Speed increases with height: $u_2/u_1 = (z_2/z_1)^p$ where $p$ = 0.1-0.4 depends on stability and roughness (power law).

## Wind rose diagram

A **wind rose** is a graphical plot of the frequency of occurrence of wind from each direction (and speed class) at a location over a period (month, season, year).

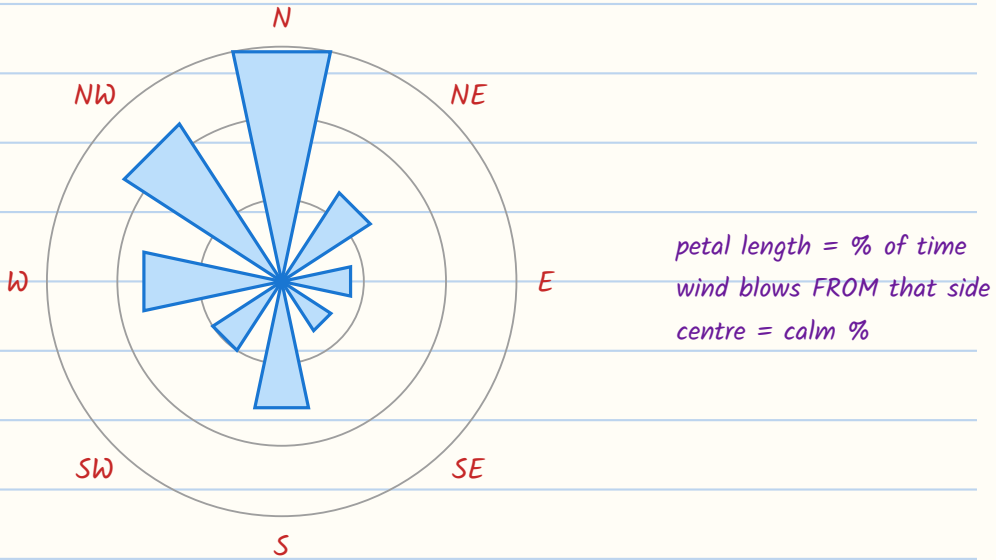


Fig. Typical wind rose (16 or 8 points)

- The length of each spoke is proportional to the % of time the wind blows FROM that direction; the longest petal is the **prevailing wind direction**
- Different widths / shades on a spoke show wind speed classes; the circle at centre shows % calm (< 1 m/s)
- **Uses:** locating industries (on the downwind side of residential areas, with a green belt), siting air monitoring stations, planning airports and roads, choosing stack positions
- Example: if prevailing wind is from SW, the residential colony must be placed to the SW of the factory (upwind).

## 4. Lapse Rates

**Lapse rate** is the rate of decrease of air temperature with increase of altitude (dT/dz). Normally temperature falls in the troposphere.

| Term | Meaning | Value |
|---|---|---|
| ELR (environmental / ambient lapse rate) | Actual temperature change of the surrounding air | Varies; mean 6.5 deg C/km |
| DALR (dry adiabatic lapse rate) | Cooling of a rising dry parcel (no heat exchange) | 9.8 deg C/km (about 1 deg C/100 m) |
| SALR (saturated adiabatic) | Cooling of a rising saturated parcel (latent heat released) | 4-6 deg C/km |
| Inversion | Temperature increases with height (ELR negative) | dT/dz > 0 |

A rising air parcel expands, does work and cools adiabatically: $T_2 / T_1 = (P_2 / P_1)^{0.286}$. Its motion depends on how its temperature compares with the surrounding air - hence stability.

## Stability conditions

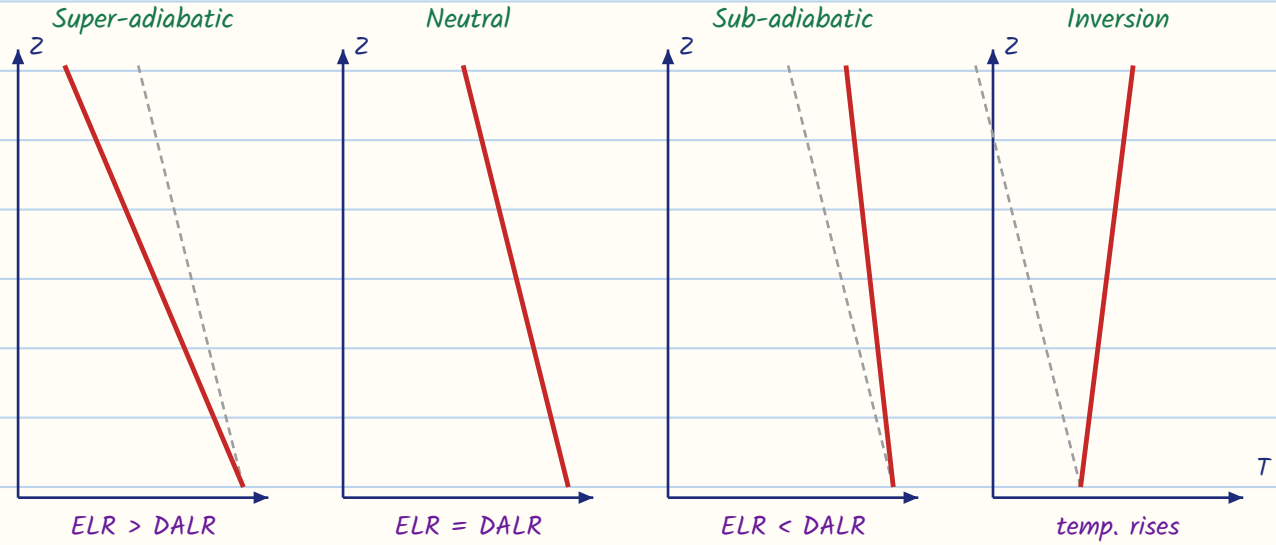



Fig. ELR (red) compared with DALR (dashed grey)

| Condition | ELR vs DALR | Behaviour of parcel | Dispersion |
|---|---|---|---|
| Unstable (super-adiabatic) | ELR > 9.8 | Displaced parcel keeps rising (warmer than surroundings) | Very good vertical mixing |
| Neutral | ELR = 9.8 | Parcel stays where it is placed | Moderate |
| Stable (sub-adiabatic) | ELR < 9.8 | Parcel returns to original level | Poor, suppressed mixing |
| Inversion | Temperature increases | Strongly stable lid | Very poor, pollutants trapped |

## Temperature inversions

- **Radiation (surface) inversion:** clear calm night, ground cools by radiation and cools the air above; breaks up after sunrise. Common in winter - morning smog
- **Subsidence inversion:** large-scale sinking of air in high-pressure region (anticyclone); air warms by compression; at 500-1500 m; can last for

days (Los Angeles)

- **Frontal inversion:** warm air overrides a cold air mass at a front
- **Advection inversion:** warm air moves over a cold surface (sea breeze over cold sea / snow)
- **Valley inversion:** cold air drains to the valley floor at night

**Inversion = lid on the atmosphere. Pollutants cannot rise above it -> high ground-level concentration (e.g. Donora 1948, London 1952, Bhopal 1984).**

## 5. Mixing Depth (Mixing Height)

**Mixing depth (MD)** is the height above the ground up to which vertical mixing of pollutants takes place by thermal turbulence. Above this height air is stable and acts as a lid.

- Determined graphically: plot the temperature-height profile (radiosonde) of early morning and draw a **dry adiabat** from the **maximum surface temperature of the afternoon**; the height where the dry adiabat meets the morning profile is the **maximum mixing depth**
- Typical values: afternoon 1000-3000 m (summer), 300-1000 m (winter), night / early morning 50-300 m
- **Ventilation coefficient (VC) = Mixing depth x mean wind speed in the mixed layer** ($m^2/s$)
- VC < 6000 $m^2/s$ -> poor dispersion (high pollution potential); VC > 10,000 $m^2/s$ -> good dispersion

**Atmospheric stability classes (Pasquill-Gifford)** - A to F:

| Surface wind (m/s) | Strong | Moderate | Slight | Night >= 4/8 cloud | Night <= 3/8 cloud |
|---|---|---|---|---|---|
| < 2 | A | A-B | B | - | - |

| Surface wind (m/s) | Strong | Moderate | Slight | Night >= 4/8 cloud | Night <= 3/8 cloud |
|---|---|---|---|---|---|
| 2 - 3 | A-B | B | C | E | F |
| 3 - 5 | B | B-C | C | D | E |
| 5 - 6 | C | C-D | D | D | D |
| > 6 | C | D | D | D | D |

A = extremely unstable, B = moderately unstable, C = slightly unstable, **D = neutral**, E = slightly stable, F = moderately stable (columns under 'strong/moderate/slight' are day-time solar insolation; D is used for overcast day or night).

## 6. Plume Behaviour (Types of Plumes)

The shape of the plume from a chimney depends on the vertical temperature profile (stability).

| Plume type | Atmospheric condition | Features |
|---|---|---|
| Looping | Strongly unstable (super-adiabatic) | Large eddies, wavy; high conc. near stack at times; good dilution |
| Coning | Neutral (about adiabatic), cloudy and windy | Cone shape; spreads at equal rate up and down |
| Fanning | Strong inversion (stable) from ground to above stack | Spreads horizontally only, travels far, no vertical mixing |
| Lofting | Inversion below, unstable above the stack | Plume goes upward - favourable, no ground effect |
| Fumigation | Unstable below, inversion above stack | Pollutants brought to the ground - worst case, high GLC |
| Trapping | Inversion above and below stack height | Plume confined between two layers; ground level high |

**Best: lofting. Worst: fumigation (e.g. at sunrise when the nocturnal inversion breaks up) and trapping.**

## 7. Plume Rise

The **effective stack height H = $h_s$ + Dh**, where $h_s$ = physical stack height and Dh = plume rise above the stack top. Plume rise occurs because of (i) **momentum** of the exit gases (velocity) and (ii) **buoyancy** (hot gases lighter than air).

Plume rise increases with: higher exit velocity $V_s$, higher stack diameter d, larger gas temperature difference ($T_s$ - $T_a$), lower wind speed u and neutral / unstable atmosphere.

### Holland's formula

- $Dh = (V_s\, d / u)\, [\, 1.5 + 2.68 \times 10^{-3}\, P\, d\, (T_s - T_a) / T_s\, ]$
- $V_s$ = stack exit velocity (m/s), d = inside diameter of stack tip (m), u = wind speed at stack height (m/s), P = atmospheric pressure (millibar), $T_s$ = stack gas temp (K), $T_a$ = air temp (K)
- For stable conditions reduce by 10-20%, for unstable increase by 10-20% (Holland's correction)

### Briggs' formula (buoyancy-dominated)

- Buoyancy flux: $F = g\, V_s\, d^2\, (T_s - T_a) / (4\, T_s)$ $(m^4/s^3)$
- Final rise (neutral/unstable): for F < 55: $Dh = 21.425\, F^{0.75} / u$; for F >= 55: $Dh = 38.71\, F^{0.6} / u$
- Gradual rise at distance x: $Dh = 1.6\, F^{1/3}\, x^{2/3} / u$

### Other formulae

- Concawe, Davidson-Bryant, Moses-Carson formulae are other empirical relations used for different conditions
- Plume rise is usually **underestimated** in calm / stable air and overestimated in strong winds - use judgement

**Numerical idea:** $V_s$ = 15 m/s, d = 3 m, u = 5 m/s, $T_s$ = 400 K, $T_a$ = 300 K, P = 1000 mb. Dh = (15 x 3 / 5)[1.5 + 2.68e-3 x 1000 x 3 x 100/400] = 9 x [1.5 + 2.01] = **31.6 m**.

## 8. Prediction of Air Quality - Dispersion Models

**Air quality models** are mathematical tools to predict the concentration of pollutants at receptors for known emission and meteorology. Uses: planning (siting), impact assessment (EIA), design of stack height, control strategy, forecasting.

- **Gaussian plume models** (point source, steady state) - ISC3, AERMOD, CALPUFF
- **Box models** (area source), **Eulerian** grid models, **Lagrangian** trajectory models
- **Statistical / receptor models** (CMB) and physical (wind tunnel) models

### 8.1 Box model

Simplest model for an **urban area source**. The city is treated as a box of length L (along wind), width W, with a lid at mixing height H.

- **Assumptions:** (1) uniform emission over the area (2) pollutants are completely and uniformly mixed within the box up to H (3) wind direction constant, wind speed u uniform (4) **steady state** (5) pollutants are conservative - no chemical reaction or removal, (6) no

pollutant crosses the top lid

- Mass balance (inflow + emission = outflow): $C_b$ u W H + q L W = C u W H
- **$C = C_b + q\ L / (u\ H)$** where q = emission rate per unit area ($g/m^2.s$), $C_b$ = background (upwind) conc.

**Example:** q = 2 x $10^{-5}$ $g/m^2.s$, L = 10 km, u = 4 m/s, H = 500 m -> C = 2e-5 x 10000 / (4 x 500) = 1 x $10^{-4}$ $g/m^3$ = **100 $ug/m^3$** (above background).

**Box model: C ~ L / (u H). Increases with larger city size and decreases with higher wind and deeper mixing. Poor for local (short distance) predictions.**

## 8.2 Gaussian plume model

Most widely used model for **continuous point sources** (chimney). Assumes the concentration distribution in the crosswind (y) and vertical (z) directions is **Gaussian (normal)** with standard deviations $sigma_y$ and $sigma_z$. The origin is at the ground below the stack, x is downwind.

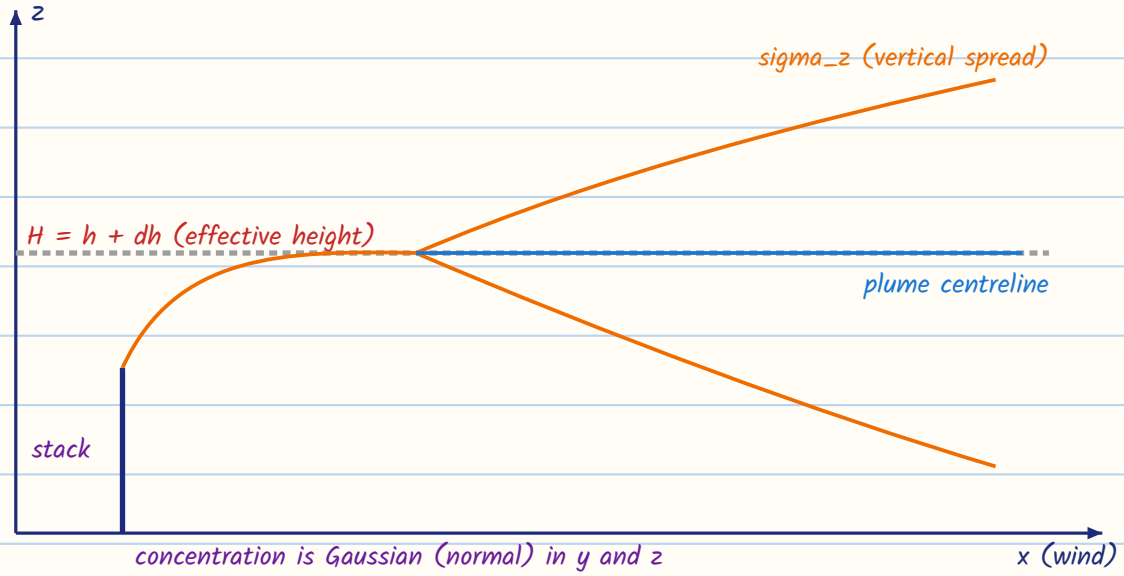



Fig. Gaussian plume - coordinate system and effective stack height

**Assumptions:** continuous steady emission; uniform wind u along x (constant with height); conservative pollutant (no reaction, decay or deposition); total reflection at ground (no absorption); flat open terrain; turbulent diffusion in x direction is negligible compared to advection; $\sigma_y$, $\sigma_z$ depend only on x and stability.

$$C(x,y,z) = Q / (2\pi u \sigma_y \sigma_z) \times \exp(-y^2 / 2\sigma_y^2) \times [\exp(-(z-H)^2 / 2\sigma_z^2) + \exp(-(z+H)^2 / 2\sigma_z^2)]$$

- C = concentration ($g/m^3$), Q = emission rate (g/s), u = mean wind speed at stack height (m/s), H = effective stack height, $\sigma_y$, $\sigma_z$ = dispersion coefficients (m)
- The second exponential term is the **image source** that accounts for ground reflection

## Special cases

- **Ground level concentration (z = 0):** $C(x,y,0) = Q / (\pi u \sigma_y \sigma_z) \exp[-y^2/2\sigma_y^2 - H^2/2\sigma_z^2]$
- **Ground level centre line (y = 0, z = 0):** $C(x,0,0) = Q / (\pi u \sigma_y \sigma_z) \exp[-H^2/2\sigma_z^2]$
- **Ground-level source (H = 0):** $C(x,0,0) = Q / (\pi u \sigma_y \sigma_z)$
- **Maximum ground level concentration:** occurs where **$\sigma_z = H / \sqrt{2}$**, and $C_{max} = 2Q / (e\pi u H^2) \times (\sigma_z / \sigma_y) \sim 0.234 Q / (u H^2)$ when $\sigma_y = \sigma_z$
- With a **mixing lid**, multiple reflections between ground and lid are added when $\sigma_z > 0.8 L$ (beyond that distance the plume is uniform in z: $C = Q / (\sqrt{2\pi}\, u \sigma_y L) \exp(-y^2/2\sigma_y^2)$)

**$C_{max}$ is inversely proportional to $H^2$: doubling the effective stack height reduces maximum GLC to about 1/4. (and u: $C \sim 1/u$)**

## 8.3 Dispersion coefficients ($\sigma_y$, $\sigma_z$)

Measure of the plume spread; they **increase with downwind distance** and **with instability**. Standard sources:

- **Pasquill-Gifford (Turner) curves** - $\sigma_y$ and $\sigma_z$ vs x for stability classes A-F (x from 0.1 to 100 km)
- **Briggs' formulae** (open country): class A: $\sigma_y = 0.22\ x\ (1 + 0.0001\ x)^{-1/2}$, $\sigma_z = 0.20\ x$; class D: $\sigma_y = 0.08\ x\ (1 + 0.0001\ x)^{-1/2}$, $\sigma_z = 0.06\ x\ (1 + 0.0015\ x)^{-1/2}$; class F: $\sigma_y = 0.04\ x\ (1 + 0.0001\ x)^{-1/2}$, $\sigma_z = 0.016\ x\ (1 + 0.0003\ x)^{-1}$ (x in metres)
- **Power law fit:** $\sigma_y = a\ x^{0.894}$; $\sigma_z = c\ x^{d} + f$ (Martin 1976) with constants a, c, d, f depending on class and x (km)
- Urban conditions have more roughness -> larger sigma (McElroy-Pooler, Briggs urban formulae)

**Procedure to solve a problem:** (1) find stability class from table (2) find effective stack height $H = h_s + \Delta h$ (3) read $\sigma_y$, $\sigma_z$ at the required x (4) substitute in the Gaussian equation (5) convert $g/m^3$ to $ug/m^3$ ($\times 10^6$).

**Example:** Q = 100 g/s, u = 4 m/s, H = 60 m, at x = 500 m (class C, Briggs): $\sigma_y = 54$ m, $\sigma_z = 38$ m. $C(500,0,0) = 100 / (3.14 \times 4 \times 54 \times 38) \times \exp(-3600 / (2 \times 1444)) = 0.00388 \times \exp(-1.25) = 0.00388 \times 0.287 = 1.1 \times 10^{-3}\ g/m^3$ = **about 1100 $ug/m^3$**.

# 9. Application of Tall Chimney for Pollutant Dispersion

Instead of treating gases, the cheapest way historically has been the **'dilution' by tall stack**: release the pollutants at a high level so that they are dispersed over a very large area and the ground-level concentration (GLC) is low.

- GLC ~ $1/H^2$: a taller chimney drastically reduces the **maximum GLC** and shifts the point of maximum farther downwind
- Takes the plume above the **inversion/ building wake zone** and avoids **downwash** (aerodynamic) and **fumigation**
- **Stack height = $h_s$.** Effective height $H = h_s + Dh$; exit velocity should be at least **1.5 times** wind speed to prevent downwash; good practice: stack height >= **2.5 times** height of nearest building
- **CPCB stack height guidelines:** $H = 14\ Q^{0.3}$ (H in m, Q = $SO_2$ emission in kg/h) for power plants; thermal plants of >= 500 MW need 275 m; 200 - 500 MW 220 m; < 200 MW $14\ Q^{0.3}$
- Other industries: minimum stack height about 30 m as per State Pollution Control Board norms; IS 6533 gives the structural design of chimneys

## Merits and demerits

| Merits | Demerits |
|---|---|
| Reduces local GLC and nuisance | Does not reduce total emission; only transfers pollution |
| Low capital cost vs control equipment | Pollutants travel far - acid rain, transboundary effects |
| Simple operation, no energy penalty | Very high structure cost, aviation hazard, may still fumigate |
| Takes plume above inversion in many cases | Not acceptable as sole control under present standards |

**Modern rule: Use tall stacks AFTER installing proper control equipment, not instead of it. 'Dilution is NOT the solution to pollution'.**

## 10. Quick Revision / Important Questions

1. Explain the factors influencing air pollution (meteorological and topographical).
2. What is a wind rose? How is it used in industrial siting?
3. Define lapse rate; explain ELR, DALR and the conditions of atmospheric stability.
4. Explain different types of plumes with neat sketches.
5. What is mixing depth? Define ventilation coefficient.
6. Derive the Gaussian plume equation; list assumptions. Derive $C_{max}$ and its location.
7. Explain the box model with assumptions. Write notes on dispersion coefficients.
8. Discuss the use and limitations of tall stacks. Explain Holland's and Briggs' plume rise formulae.

# AIR POLLUTION AND CONTROL

## Subject Code : 23A01504b

## UNIT - III : CONTROL OF PARTICULATE POLLUTANTS

**Syllabus:** Properties of particulate pollution - Particle size distribution - Control mechanism - Dust removal equipment - Design and operation of settling chambers, cyclones, wet dust scrubbers, fabric filters and ESP.

## I. Properties of Particulate Pollutants

The choice and design of control equipment depends on the **properties of the particles** and of the carrier gas.

### I.I Particle properties

- **Size (diameter):** most important property; ranges from 0.001 um to > 100 um. For irregular particles an equivalent diameter is used: aerodynamic, Stokes, projected-area diameter
- **Aerodynamic diameter:** diameter of a unit-density (1 $g/cm^3$) sphere having the same settling velocity as the particle
- **Shape:** spherical, fibrous, flakes, irregular; non-spherical particles settle slower (shape factor)
- **Density:** true (particle) density and bulk density; settling velocity increases with density
- **Specific surface:** surface area per unit mass - high for fine particles; adsorbs gases, high reactivity
- **Cohesion / adhesion:** sticky or hygroscopic dust blocks filters and sticks to surfaces; very fine dust agglomerates
- **Electrical properties:** charge acquired and **electrical resistivity** (important for ESP)

- **Optical properties:** scatter light (visibility, opacity) - maximum scattering for 0.1 - 1 um
- **Hygroscopicity, flammability, explosivity, toxicity, abrasiveness, corrosiveness** - influence the material of construction and safety

### 1.2 Gas properties

- Temperature, pressure, humidity, **dew point**, viscosity, density, flow rate, composition (acid gases), dust loading ($g/m^3$)

| Size range (um) | Behaviour in air | Typical particles |
|---|---|---|
| > 10 | Settle fast by gravity; removed in nose | Dust, sand, fly ash coarse |
| 1 - 10 | Settle slowly; impaction | Fly ash, cement, pollen |
| 0.1 - 1 | Stay airborne for days; best light scattering | Smoke, fumes, acid mist |
| < 0.1 | Brownian motion, diffusion; coagulate | Fumes, combustion nuclei |

## 2. Particle Size Distribution

A dust sample contains particles of many sizes. Its **particle size distribution** shows the fraction of mass (or number) in each size range. It is measured by sieving, microscopy, sedimentation, cascade impactor, laser diffraction, coulter counter etc.

- **Frequency distribution:** % mass in each size interval (histogram)
- **Cumulative distribution:** % mass **smaller than** (undersize) or **larger than** (oversize) a given size; plotted on **log-probability paper**
- Most air-borne dust follows a **log-normal distribution** - a straight line on log-probability graph

### Important parameters

- **Mass median diameter (MMD or $d_{50}$):** diameter above and below which 50% of the mass lies (read at 50% on the cumulative plot)
- **Geometric mean diameter:** $d_g = (d_1^{n1} \times d_2^{n2} \times \dots)^{1/N}$ or $\log d_g = \text{sum}(n_i \log d_i) / N$
- **Geometric standard deviation:** $\text{sigma}_g = d_{84.1} / d_{50} = d_{50} / d_{15.9}$ ($\text{sigma}_g = 1$ for monodisperse, > 1.2 polydisperse)
- **Number average diameter** $d_n = \text{sum}(n_i d_i) / \text{sum}(n_i)$; **surface mean** and **volume mean** diameters
- **Hatch-Choate relations** connect the different mean diameters of a log-normal distribution: $\text{MMD} = \text{CMD} \times \exp(3 \ln^2 \text{sigma}_g)$ (CMD = count median diameter)

**Why needed? Collection efficiency of every device is a function of particle size - we need a fractional efficiency curve and the size distribution to find the overall efficiency: $\text{eta}_o = \text{sum}(\text{eta}_i \times m_i)$ where $m_i$ = mass fraction of size i.**

**Example:** three fractions with mass 20%, 50%, 30% and collection efficiencies of 40%, 80%, 95% -> $\text{eta}_o = 0.2 \times 40 + 0.5 \times 80 + 0.3 \times 95 = 8 + 40 + 28.5 =$ **76.5%**.

## 3. Control Mechanisms (Collection Mechanisms)

Particles are separated from the gas stream by applying a force that moves them to a collecting surface.

| Mechanism | Principle | Used in / best for |
|---|---|---|
| Gravitational settling | Gravity force on particle; $v_s$ | Settling chamber; > 50 um |
| Centrifugal force | Spinning gas throws particles to wall | Cyclone; > 5-10 um |

| Mechanism | Principle | Used in / best for |
|---|---|---|
| Inertial impaction | Particle cannot follow gas streamline around obstacle (droplet, fibre) | Scrubbers, filters; > 1 um |
| Interception | Particle following streamline touches the collector | Filters, scrubbers; 1 um |
| Diffusion (Brownian) | Random motion brings very small particles to the surface | Filters, scrubbers; < 0.3 um |
| Electrostatic attraction | Charged particle moves in electric field to electrode | ESP; 0.1 - 10 um |
| Thermophoresis, sonic agglomeration, condensation | Auxiliary mechanisms | Special cases |

## Stokes' law and settling velocity

$v_s = g\, d^2\, (rho_p - rho_g) / (18\, mu)$ [valid for Re < 1, d in m]

- $v_s$ = terminal settling velocity (m/s), d = particle diameter (m), $rho_p$ = particle density, $rho_g$ = gas density, mu = gas viscosity (air $1.8 \times 10^{-5}$ kg/m.s)
- $v_s$ proportional to $d^2$: a 10 um particle settles 100 times faster than a 1 um one
- For very small particles (< 1 um) use **Cunningham slip correction factor** $C_c > 1$: $v_s = C_c\, g\, d^2\, rho_p / 18\, mu$
- For Re 1 to 1000 (intermediate) use $v_s = [4\, g\, d\, (rho_p - rho_g) / (3\, C_D\, rho_g)]^{1/2}$ with $C_D = 24/Re \times (1 + 0.15\, Re^{0.687})$

**Example:** d = 50 um, $rho_p$ = 2000 kg/m$^3$: $v_s = 9.81 \times (50 \times 10^{-6})^2 \times 2000 / (18 \times 1.8 \times 10^{-5}) = 4.9 \times 10^{-5} / 3.24 \times 10^{-4}$ = **0.15 m/s**.

## 4. Dust Removal Equipment - Overview

| Equipment | Mechanism | Size removed | Efficiency | Pressure drop |
|---|---|---|---|---|
| Settling chamber | Gravity | > 50 um | 40 - 60% | 5 - 25 Pa |
| Cyclone | Centrifugal | > 5 um | 70 - 90% | 0.5 - 1.5 kPa |
| Wet scrubber (venturi) | Impaction, diffusion | > 0.5 um | 90 - 99% | 2 - 10 kPa |
| Fabric filter | Sieving, impaction, diffusion | > 0.1 um | 99 - 99.9% | 1 - 2 kPa |
| ESP | Electrostatic | > 0.1 um | 95 - 99.9% | 0.1 - 0.5 kPa |

**Selection depends on:** dust size and concentration, required efficiency, gas flow, temperature, corrosiveness, cost (capital + operating), space, availability of water, disposal of collected dust. Equipment are often used in series: **cyclone (pre-cleaner) + ESP / fabric filter (final cleaner)**.

## 5. Gravity Settling Chamber

A large rectangular chamber in which the gas velocity is reduced (1 - 3 m/s -> 0.3 m/s) so that the particles have enough time to settle. Used as **pre-cleaner** for coarse dust.

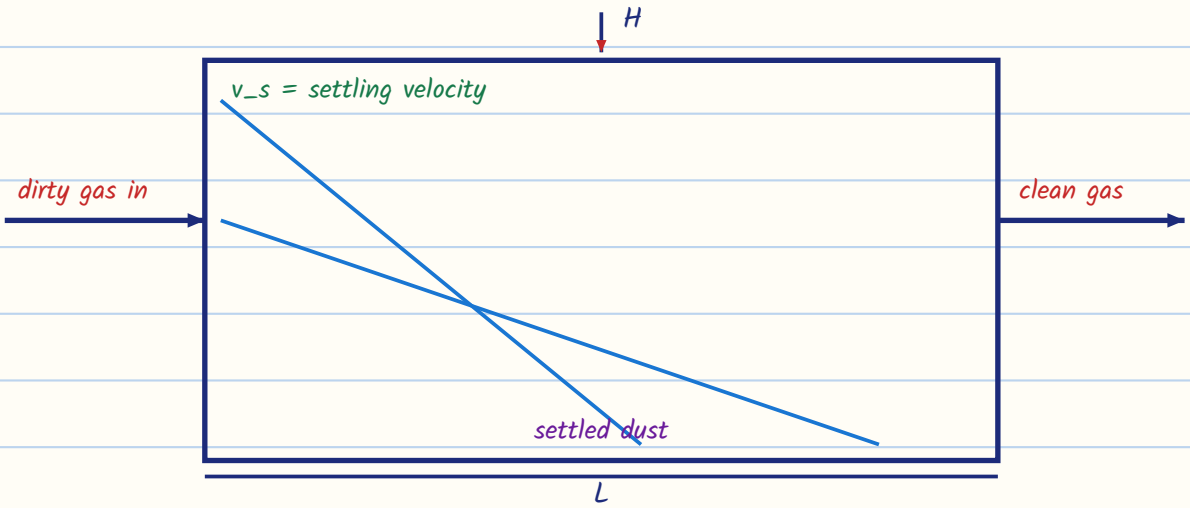



Fig. Horizontal-flow gravity settling chamber

## Design theory (laminar flow model)

- Residence time of gas: $t = L / v_h$ ($v_h$ = horizontal velocity = $Q / (W \times H)$)
- A particle at the top must fall height H in that time: $H / v_s <= L / v_h$
- **Efficiency (laminar / ideal, one size): $\eta = v_s L / (v_h H) = v_s L W / Q$** (100% when $v_s L / v_h H >= 1$)
- **Turbulent mixing model: $\eta = 1 - \exp( - v_s L W / Q )$**
- **Minimum size completely removed (cut size): $d_{min} = [ 18 \mu Q / (g \rho_p W L) ]^{1/2}$**
- Efficiency depends on the **plan area (L x W)**, not on the height H: use many trays (**Howard multiple-tray settling chamber**) - n trays give n times the efficiency (but cleaning is difficult)
- Design limits: $v_h$ < 3 m/s (to avoid re-entrainment of settled dust); typical $v_h$ 0.3 - 0.6 m/s; Re < 2000 for laminar flow; length is generally 2-3 times width
- Pressure drop very small; only for particles > 50 um (40-60% overall efficiency)

| Advantages | Disadvantages |
|---|---|
| Simple, cheap, no moving parts | Huge size / space needed |
| Low pressure drop, easy to maintain | Only coarse particles; low efficiency |
| Can handle high temperature and corrosive gases | Difficult to clean; re-entrainment |

**Example:** $Q = 1.5\ m^3/s$, $W = 3$ m, $L = 6$ m, $d = 50$ um ($v_s = 0.15$ m/s): $\eta = 0.15 \times 6 \times 3 / 1.5 = 1.8 > 1$, so **100%**; for $d = 20$ um ($v_s = 0.024$ m/s): $\eta = 0.024 \times 18 / 1.5 =$ **29%**.

# 6. Cyclone Separators

**Centrifugal force** (many times gravity) is used. Gas enters **tangentially** at the top of a cylindrical-conical vessel, spirals down as the **outer vortex**; particles are thrown to the wall, slide down into the hopper; the clean gas reverses and leaves as the **inner vortex** through the central outlet (vortex finder).

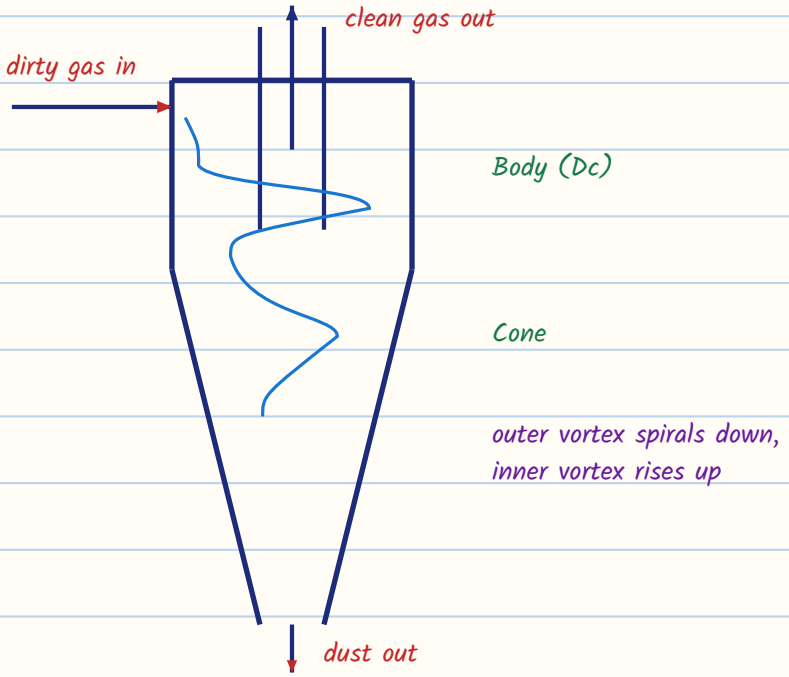



Fig. Reverse-flow (tangential inlet) cyclone

## Standard (Lapple) cyclone dimensions

| Symbol | Part | Dimension |
|---|---|---|
| $D_c$ | Body diameter | $D_c$ |
| $H_c$ | Inlet height (a) | $D_c / 2$ |
| $B_c$ | Inlet width (b) | $D_c / 4$ |
| $D_e$ | Gas exit (vortex finder) diameter | $D_c / 2$ |
| $S_c$ | Length of vortex finder | $D_c / 8$ |
| $L_b$ | Cylinder length | $2\ D_c$ |
| $L_c$ | Cone length | $2\ D_c$ |
| $J_c$ | Dust outlet diameter | $D_c / 4$ |

## Design equations

- Inlet velocity $V_i = Q / (H_c \times B_c)$; good practice **15 - 25 m/s** (below 15 poor efficiency; above 25 re-entrainment/erosion)
- **Number of effective turns:** $N = (1 / H_c)(L_b + L_c / 2)$ = about 5 for Lapple standard cyclone
- **Cut diameter (50% efficiency):** $d_{50} = [9 \, mu \, B_c / (2 \, pi \, N \, V_i \, (rho_p - rho_g))]^{1/2}$
- **Lapple fractional efficiency:** $eta_i = 1 / [1 + (d_{50} / d_i)^2]$
- **Pressure drop:** $dP = (rho_g \, V_i^2 / 2) \times K_c$, $K_c = 16 \, H_c \, B_c / D_e^2$ (about 8 velocity heads for standard design)
- **Overall efficiency** = sum of (mass fraction x fractional efficiency)

**Example:** $D_c = 0.5$ m, $Q = 0.5$ m$^3$/s -> $H_c = 0.25$, $B_c = 0.125$, $V_i = 0.5 / 0.03125 = 16$ m/s. With $N = 5$, $mu = 1.8 \times 10^{-5}$, $rho_p = 2000$: $d_{50} = sqrt(9 \times 1.8e\text{-}5 \times 0.125 / (2 \, pi \times 5 \times 16 \times 2000))$ = **4.5 um**; $dP = 0.5 \times 1.2 \times 16^2 \times 8$ = **1230 Pa**.

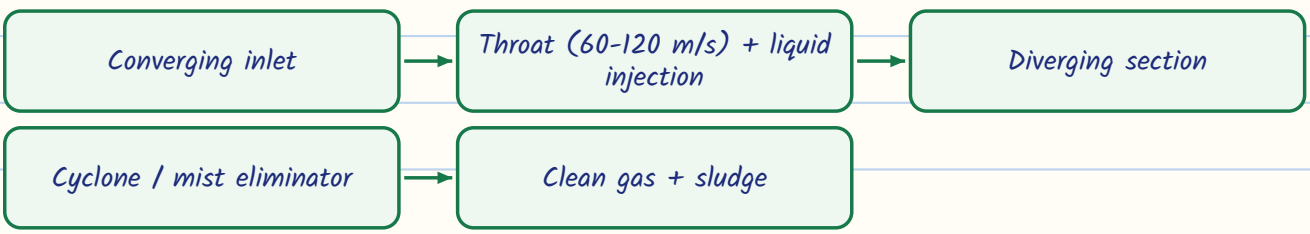
- **High-efficiency cyclones** (Stairmand) are tall and narrow with smaller inlet; **high-throughput** types have larger inlet; **multicyclones** (many small tubes in parallel) give 90% for 5 um
- Variants: cyclones with axial (straight-through) entry, reverse-flow, dust-return; **dry** and **wet (irrigated)**
- **Advantages:** cheap, no moving parts, compact, handles high temperature and pressure and heavy dust loads (up to 1000 g/m$^3$), low maintenance
- **Disadvantages:** poor for particles < 5 um, efficiency falls at low load, erosion in abrasive dust, high pressure drop and cannot handle sticky material

# 7. Wet Dust Scrubbers (Wet Collectors)

The dusty gas is brought into **intimate contact with liquid (water) droplets** or films; particles are captured by droplets (**inertial impaction, interception, diffusion**), then the droplets are removed by a mist eliminator. Capture of particles by impaction is given by the **target efficiency**, which rises with particle size, velocity and with small droplet size.

| Type | Working | Remarks |
|---|---|---|
| Spray tower | Water sprayed down, gas up; droplets 500-1000 um | Low dP (0.1-0.5 kPa); > 10 um only |
| Cyclone spray scrubber | Cyclone with water spray / wall film | Higher efficiency, > 2 um |
| Impingement plate / tray | Gas passes through holes onto plates, water on tray; froth | > 1 um; 1-2 kPa |
| Packed bed (tower) | Random packing wetted with liquid, counter/cross flow | For gas absorption + clean dust; clogging problem |
| Venturi scrubber | Gas accelerated in throat 60-120 m/s, liquid injected; atomised | Best for submicron; 2.5-15 kPa |
| Mechanical (rotor) scrubber | Power-driven rotor gives fine spray | High cost, energy use |

## Venturi scrubber




- Liquid : gas ratio **0.7 - 2 L / $m^3$** of gas; pressure drop 2.5 - 15 kPa, efficiency > 99% for 0.5 um dust at high dP

- **Johnstone equation: eta = 1 - exp( - k R sqrt(psi) )**; R = liquid / gas ratio ($m^3$ / $m^3$), k = correlation constant (0.1 - 0.2), **inertial impaction parameter psi = C rho$_p$ v$_t$ d$_p^2$ / (18 mu d$_0$)** where v$_t$ = throat velocity, d$_0$ = droplet diameter
- **Pressure drop** (Calvert) increases with throat velocity squared and L/G; **power consumption** is the main cost - 'collection efficiency vs energy' relation

| Advantages | Disadvantages |
|---|---|
| Collects both gas and particulate; cools hot gases | Water pollution: sludge disposal / treatment |
| High efficiency for fine and sticky dust | High energy use (venturi); corrosion and freezing |
| No explosion risk; compact; low cost | Visible steam plume; costly maintenance |

## 8. Fabric Filters (Bag House)

The dusty gas is passed through **woven or felted fabric bags** that retain the particles. Initially the cloth itself is the filter, but very soon a **dust cake** (layer) forms on the fabric surface which acts as the main filter medium - giving high efficiency even for submicron particles.

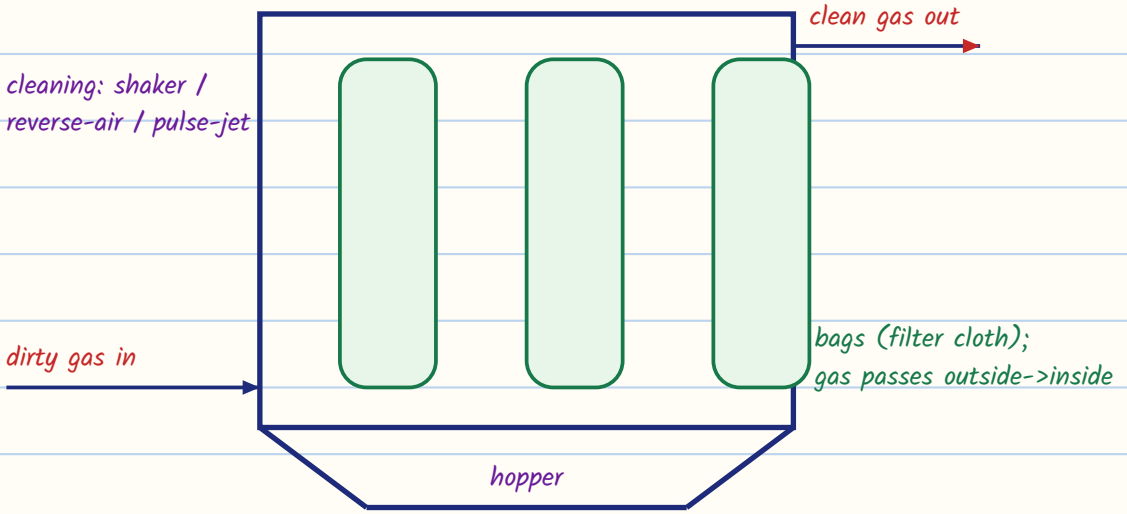


Fig. Baghouse (fabric filter) with cleaning arrangement

## Collection mechanisms

- **Sieving** by the cake (dominant), **inertial impaction** (> 1 um), **interception**, **diffusion** (< 0.3 um), **electrostatic** (charge on fibres and particles)

## Design

- **Air-to-cloth ratio (filtration velocity): $V_f = Q / A$** ($m^3$/min per $m^2$ = m/min). Typical 0.5 - 2.5 m/min (shaker 0.5-1, reverse air 0.5-1, pulse-jet 2-4)
- **Cloth area required: $A = Q / V_f$**; number of bags $n = A / (\pi D L)$ with bag diameter D = 0.1 - 0.3 m, length 3 - 10 m; add 10-20% for compartment being cleaned off-line
- **Pressure drop: $dP = dP_f + dP_c = K_1 V_f + K_2 C V_f^2 t$** ($K_1$ fabric resistance, $K_2$ specific cake resistance, C = inlet dust conc., t = filtering time) - typical 0.5 - 2 kPa (50 - 200 mm w.g.)
- Gas must be cooled above its dew point: temperature limit decided by the cloth

| Fabric | Max. temp (deg C) | Remarks |
|---|---|---|
| Cotton | 80 | Cheap; poor acid resistance |
| Wool | 100 | Good for acid; poor in alkali |
| Nylon / polyamide | 200 | Abrasion resistant; poor to acid |
| Polyester (Dacron) | 135 | Good general-purpose, poor in hot water |
| Nomex (aramid) | 220 | Good dimensional stability |
| Glass fibre | 260 | High temperature; weak to flexing |
| Teflon (PTFE) | 260 | Excellent chemical resistance; costly |

### Cleaning methods

1. **Mechanical shaking:** bag top oscillated; the filtering for that compartment is stopped. Simple; 'woven' fabrics
2. **Reverse air:** reverse gentle flow collapses the bag and cake falls; for glass fibre
3. **Pulse-jet:** short burst (0.1 s) of compressed air (5-7 bar) down the bag; **on-line** cleaning; felt; high ratio, compact

- **Advantages:** efficiency 99-99.9% for all sizes, handles variable loads, dry collection of product, simple and modular
- **Disadvantages:** limited temperature, moisture / sticky dust blinds the bag, fire hazard, bag life 1-3 years, large space for low-ratio types, high maintenance

## 9. Electrostatic Precipitator (ESP)

Uses **electrostatic forces** to separate charged particles. A very high DC voltage (40 - 100 kV) is applied between discharge electrodes (wires, negative) and grounded collecting plates, producing a **corona discharge** around the wire.

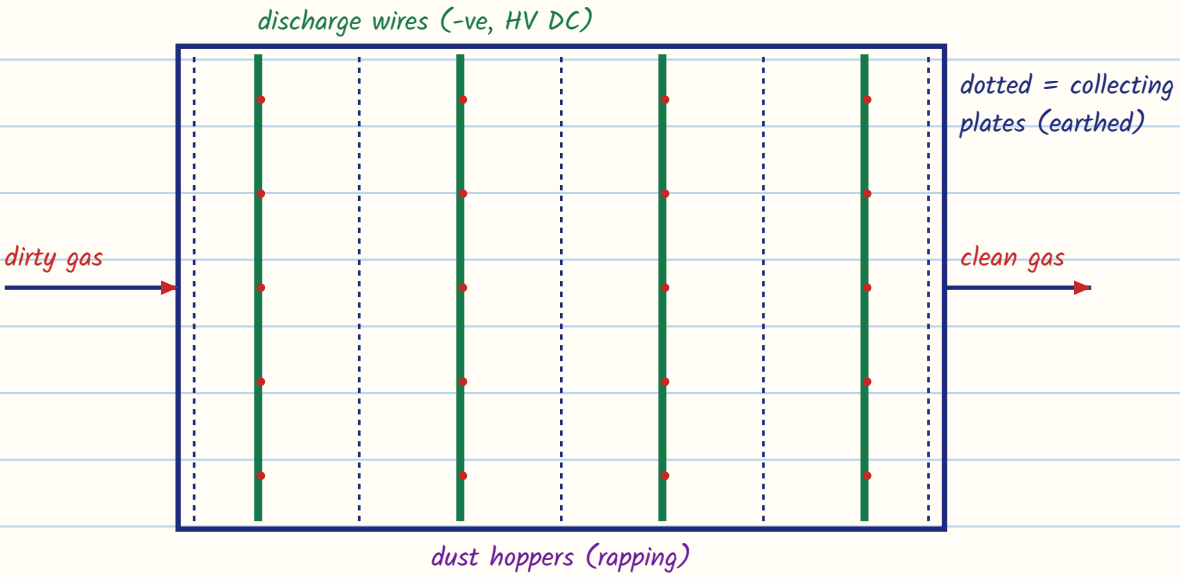



Fig. Plate-type electrostatic precipitator

## Working steps

1. **Ionisation:** corona produces gas ions (electrons and negative ions)
2. **Charging:** particles capture ions by field charging (> 1 um) and diffusion charging (< 0.2 um)
3. **Migration:** charged particles move towards the collecting electrode under electric field, drift velocity (migration velocity) w
4. **Deposition:** particles stick to the plates as a layer
5. **Removal:** plates are **rapped** (hammers / vibrators), dust falls to the hopper; for wet ESP the plates are flushed with water

## Design - Deutsch-Anderson equation

**eta = 1 - exp( - w A / Q ) = 1 - exp( - w L / (v d) ) -> A = - (Q / w) ln (1 - eta)**

- eta = collection efficiency, w = migration (drift) velocity (about 0.05 - 0.2 m/s), A = total collecting plate area ($m^2$), Q = gas flow ($m^3/s$)
- **Migration velocity: w = q $E_p$ / (3 pi mu d)** (q = particle charge, $E_p$ = field strength at collecting plate) - w proportional to field strength squared and particle size
- **Specific collecting area (SCA) = A / Q** (s/m) = about 30-80 for fly ash (A in $m^2$ per $m^3/s$); Gas velocity 1 - 2 m/s; plate spacing 20 - 40 cm; treatment time 2-10 s
- Efficiency is increased by increasing A (more fields), reducing gas velocity (Q), and improving w (voltage, conditioning)
- **Modified (Matts-Ohnfeldt)** equation: eta = 1 - exp[ - $(w A/Q)^k$ ], k = 0.4 - 0.6, accounts for non-ideal conditions: gas sneakage, re-entrainment

**Example:** Q = 50 $m^3/s$, w = 0.1 m/s, eta = 99% -> A = -(50 / 0.1) x ln(0.01) = 500 x 4.6 = **2300 $m^2$**.

### Dust resistivity (very important)

| Resistivity (ohm-cm) | Effect | Remedy |
|---|---|---|
| $< 10^4$ | Dust loses charge instantly; re-entrained (e.g. carbon) | Add ammonia, use baffles / wet ESP |
| $10^8 - 10^{10}$ | **Ideal** range for good collection | - |
| $> 10^{11}$ | **Back corona**: insulating layer, sparking, low efficiency (low-S coal fly ash) | Condition with $SO_3$, $H_2O$, $NH_3$, Na; hot-side ESP; pulse energisation |

- **Types:** plate-wire (most common), tubular, **wet** ESP, single-stage (charging + collection together) and **two-stage** (air conditioning), hot-side and cold-side
- **Advantages:** high efficiency (99.9%), low pressure drop (< 0.25 kPa), handles large flows and high temperature (up to 400 deg C), low operating cost, dry recovery of valuable dust
- **Disadvantages:** high capital cost, large space, sensitivity to resistivity and flow variations, produces ozone, not for explosive gases, needs skilled maintenance

**Applications: thermal power plants (fly ash), cement kilns, steel (sinter plants, blast furnaces), paper (recovery boilers), non-ferrous smelters.**

## 10. Quick Revision / Important Questions

1. Explain particle size distribution; define MMD and geometric standard deviation.
2. Explain the collection mechanisms used in particulate control.

3. Derive the efficiency and minimum particle size of a gravity settling chamber.
4. With a neat sketch explain the working and design of a cyclone; write the Lapple formulae.
5. Describe a venturi scrubber. Discuss merits and demerits of wet scrubbers.
6. Explain a baghouse; the cleaning methods and air-to-cloth ratio.
7. Explain the working of an ESP. Derive the Deutsch-Anderson equation. Effect of dust resistivity?

# AIR POLLUTION AND CONTROL

## Subject Code : 23A01504b

## UNIT – IV : CONTROL OF GASEOUS POLLUTANTS

**Syllabus:** Process and equipment for the removal by chemical methods - Design and operation of absorption and adsorption equipment - Combustion and condensation equipment.

## I. Introduction – Control of Gaseous Pollutants

Gaseous pollutants ($SO_2$, $NO_x$, $H_2S$, HCl, HF, $Cl_2$, $NH_3$, CO, VOCs, hydrocarbons) cannot be removed by mechanical filters. They are controlled by **mass transfer** or **chemical conversion**.

Source control (change fuel / process) → Gas cleaning (end-of-pipe) → Tall stack dispersion (last resort)

| Method | Principle | Typical use |
|---|---|---|
| Absorption (scrubbing) | Gas dissolves in liquid, physically or with reaction | $SO_2$, HCl, $NH_3$, $H_2S$, $NO_x$ |
| Adsorption | Gas molecules stick to the surface of a solid | VOC, solvent vapour, odour, Hg |
| Combustion / oxidation | Burn to $CO_2$ and $H_2O$ | VOC, CO, odours, HC |
| Condensation | Cool / compress to liquefy vapour | High-conc. solvent vapour |
| Chemical conversion (catalytic) | Convert to harmless product using catalyst | $NO_x$ (SCR), CO / HC (converter) |

| Method | Principle | Typical use |
|---|---|---|
| Biofiltration | Micro-organisms degrade pollutants | Odours, low-conc. VOC, $H_2S$ |

**Selection factors:** pollutant type and concentration, gas flow and temperature, required efficiency, recovery value, waste generated, capital and operating cost, safety (flammability), space.

## 2. Removal by Chemical Methods

In chemical methods the pollutant is converted to a harmless or valuable product by chemical reaction with a reagent (alkali, ammonia, lime, oxygen) in a reactor / scrubber.

### 2.1 Control of sulphur dioxide - Flue Gas Desulphurisation (FGD)

- **Non-regenerative (throwaway):** reagent is used once and the product is disposed or sold (gypsum)
- **Regenerative:** reagent is recovered and sulphur by-product (S, $H_2SO_4$, liquid $SO_2$) is sold (Wellman-Lord, magnesium oxide, citrate process)

| FGD process | Reagent / reaction | Efficiency |
|---|---|---|
| Wet limestone | $CaCO_3 + SO_2 + 1/2\ O_2 + 2\ H_2O \rightarrow CaSO_4.2H_2O$ (gypsum) + $CO_2$ | 90 - 98% |
| Wet lime | $Ca(OH)_2 + SO_2 \rightarrow CaSO_3.1/2\ H_2O$; oxidised to $CaSO_4$ | 90 - 98% |
| Spray dry (semi-dry) | Lime slurry atomised in hot gas; dry $CaSO_3$ collected in baghouse | 70 - 90% |
| Dry sorbent injection | Lime / sodium bicarbonate injected into duct | 30 - 60% |

| FGD process | Reagent / reaction | Efficiency |
|---|---|---|
| Double alkali | NaOH scrubbing; regenerate with lime: $Na_2SO_3$ + $Ca(OH)_2$ -> $CaSO_3$ + 2 NaOH | 90 - 95% |
| Wellman-Lord | $Na_2SO_3$ + $SO_2$ + $H_2O$ -> 2 $NaHSO_3$; heat gives conc. $SO_2$ | 90 - 97% |

- Other $SO_2$ methods: fluidised bed combustion with limestone, coal washing, switching to low-sulphur fuels, **ammonia scrubbing** (gives ammonium sulphate fertiliser)

## 2.2 Control of oxides of nitrogen ($NO_x$)

- **Combustion modification (primary):** low-$NO_x$ burners, staged combustion, flue gas recirculation, lower excess air and lower peak temperature
- **Selective Catalytic Reduction (SCR):** 4 NO + 4 $NH_3$ + $O_2$ -> 4 $N_2$ + 6 $H_2O$ at **300 - 400 deg C** over $V_2O_5/TiO_2$ catalyst; 80 - 95% removal; ammonia slip < 5 ppm
- **Selective Non-Catalytic Reduction (SNCR):** $NH_3$ or urea injected at **850 - 1100 deg C**; 30 - 60% removal; no catalyst
- **Wet scrubbing:** NO is poorly soluble; oxidise (ozone, $ClO_2$, $H_2O_2$) to $NO_2$ then absorb in alkali: 2 $NO_2$ + 2 NaOH -> $NaNO_2$ + $NaNO_3$ + $H_2O$
- **Three-way catalytic converter** (vehicles): $NO_x$ reduced, CO and HC oxidised (see Unit V)

## 2.3 Control of $H_2S$, HCl, HF, $Cl_2$, $NH_3$ and CO

- **$H_2S$:** absorbed in amine (MEA, DEA) or alkaline solutions; **Claus process** converts it to sulphur: $H_2S + 3/2\ O_2 \rightarrow SO_2 + H_2O$; $2\ H_2S + SO_2 \rightarrow 3\ S + 2\ H_2O$ ($Al_2O_3$ catalyst)
- **HCl, HF, $Cl_2$:** highly soluble - water or caustic soda scrubbers: $HCl + NaOH \rightarrow NaCl + H_2O$; $Cl_2 + 2\ NaOH \rightarrow NaCl + NaOCl + H_2O$
- **$NH_3$:** absorbed in dilute acid ($H_2SO_4$) - gives ammonium sulphate
- **CO:** oxidised over a Hopcalite ($MnO_2$-CuO) or noble metal catalyst; $2\ CO + O_2 \rightarrow 2\ CO_2$
- **VOCs:** catalytic oxidation, bio-oxidation, UV photocatalysis

# 3. Absorption

**Absorption** is the transfer of a gaseous pollutant (solute) from the gas mixture into a liquid (solvent / absorbent). Physical absorption depends on **solubility**; chemical absorption (reaction in the liquid) greatly increases the rate and capacity.

## 3.1 Fundamentals

- **Henry's law** (dilute solutions): $p = H\ x$ or $y = m\ x$, $m = H / P$ ($p$ = partial pressure of solute, $x$ = mole fraction in liquid, $H$ = Henry's constant). Low $H \rightarrow$ highly soluble gas ($NH_3$, HCl); high $H \rightarrow$ poorly soluble (CO, NO)
- **Equilibrium line** ($y^* = m\ x$) and **operating line** (mass balance) decide driving force ($y - y^*$)
- **Two-film theory (Whitman):** resistance to transfer lies in a thin gas film and a thin liquid film at the interface. $1/K_G = 1/k_g + H/k_l$. Rate $N_A = K_G\ a\ (p - p^*)$ per volume

- For **very soluble** gases the **gas film** controls ($NH_3$, HCl); for **sparingly soluble** gases the **liquid film** controls ($CO_2$, $O_2$, NO); chemical reaction removes the liquid-film resistance
- **Absorption factor A = L / (m G)**: if A > 1, high removal is easy

## 3.2 Choice of absorbent (solvent)

- High solubility for solute and **selective**; low volatility (less solvent loss); low viscosity; non-corrosive, non-toxic, non-flammable, cheap, easily regenerated, stable
- Water for $NH_3$, HCl, HF, $SO_2$ (moderate); NaOH / $Na_2CO_3$ / lime for acid gases; amines for $H_2S$ and $CO_2$; oils for hydrocarbons

## 3.3 Absorption equipment (gas scrubbers)

| Equipment | Features | Application |
|---|---|---|
| Packed tower | Gas and liquid contact over packing surface; counter-current | Most common; > 95% removal for soluble gases |
| Plate (tray) tower | Sieve / bubble-cap / valve trays; gas bubbles through liquid | Large flows, large liquid loads, fouling liquids |
| Spray tower / chamber | Liquid sprayed into rising gas; low dP | Highly soluble gases; combined with PM removal |
| Venturi scrubber | High velocity atomises liquid | Simultaneous particulate + gas |
| Bubble column | Gas bubbled through liquid pool | Reactive gases, slurry |

## 3.4 Packed tower

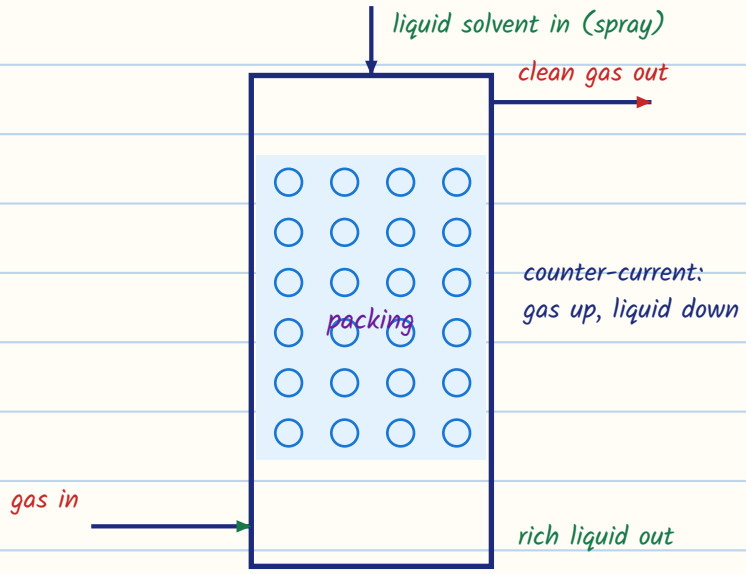



Fig. Counter-current packed absorption tower

- **Shell** (steel / FRP / PVC) filled with **packing** supported on a grid; **liquid distributor** at the top gives uniform wetting; **gas distributor** at the bottom; **mist eliminator** on top; **redistributors** every 3 - 5 tower diameters
- **Packing** gives large surface area, voidage, low pressure drop: **random** - Raschig rings, Berl saddles, Intalox saddles, **Pall rings** (best), Tellerettes; **structured** - corrugated sheets (low dP, costly); materials: ceramic, plastic, metal, carbon
- Packing size: 1/8 th of tower diameter at most; typical size 25 - 50 mm

### Design of a packed tower (dilute gas, straight equilibrium line)

1. **Material balance:** $G\,(y_1 - y_2) = L\,(x_1 - x_2)$ (G and L = molar flows of inert gas and pure solvent; use Y and X ratios for non-dilute)
2. **Minimum liquid rate:** $(L/G)_{min} = (y_1 - y_2) / (x_1^* - x_2)$ with $x_1^* = y_1/m$; **actual** $L = 1.2 - 1.5 \times L_{min}$

3. **Tower diameter:** from gas flow and permissible superficial velocity = **50 - 75% of flooding velocity** (from Sherwood-Leva-Eckert generalised pressure drop chart); D = sqrt( 4 Q / (pi v) ); pressure drop 200 - 400 Pa per metre of packing
4. **Height of packing: Z = $H_{OG}$ x $N_{OG}$** (height of a transfer unit x number of transfer units)
5. **$H_{OG}$ = G / ($K_y$ a)** (typically 0.3 - 1 m); **$N_{OG}$ = integral dy / (y - $y^*$)** from $y_2$ to $y_1$
6. For a dilute gas with straight lines (Colburn): **$N_{OG}$ = [ 1 / (1 - 1/A) ] ln{ (1 - 1/A) ($y_1$ - m $x_2$) / ($y_2$ - m $x_2$) + 1/A }**, with A = L / (mG)
7. If $x_2$ = 0 and A very large, $N_{OG}$ = ln ($y_1$ / $y_2$) = ln (1 / (1 - eta))

**Example:** to remove 99% of a solute with $N_{OG}$ = ln (100) = 4.6 and $H_{OG}$ = 0.6 m -> **Z = 4.6 x 0.6 = 2.8 m** (plus 15-20% safety).

**Packed tower: high contact area, low dP, suitable for corrosive gas (plastic) | Plate tower: for large L/G, solids, easy cleaning and cooling coils.**

- **Operating problems:** flooding (excess gas / liquid), channelling, foaming, fouling and scaling ($CaSO_4$), loading of dust, **weeping** and dumping of trays
- **Disposal:** spent solution needs treatment (neutralise, settle); some gives by-products (gypsum, ammonium sulphate)

## 4. Adsorption

**Adsorption** is the accumulation (adhesion) of molecules of a gas (**adsorbate**) on the surface of a solid (**adsorbent**). It is a surface phenomenon (different from absorption) and is exothermic.

| Property | Physical adsorption (physisorption) | Chemical adsorption (chemisorption) |
|---|---|---|
| Forces | Van der Waals (weak) | Chemical bonds (strong) |
| Heat of adsorption | 10 - 40 kJ/mol | 40 - 400 kJ/mol |
| Reversibility | Reversible; easy regeneration | Irreversible mostly |
| Layers | Multilayer | Monolayer |
| Temperature | Decreases with temperature rise | Increases first, then decreases |
| Specificity | Not specific | Highly specific |

## 4.1 Adsorbents

| Adsorbent | Surface area ($m^2/g$) | Application |
|---|---|---|
| Activated carbon (coal / coconut shell) | 600 - 1500 | Non-polar VOCs, solvents, odours, Hg, $H_2S$, dioxins |
| Silica gel | 200 - 800 | Dehydration of gases, polar compounds |
| Activated alumina | 200 - 350 | Moisture, HF, $H_2S$, drying |
| Molecular sieves / zeolites | 500 - 800 | Selective (pore size 3A, 4A, 5A, 13X), drying, $SO_2$, $CO_2$ |
| Polymeric resin, fly ash, lime | Variable | Special organics, acid gases |

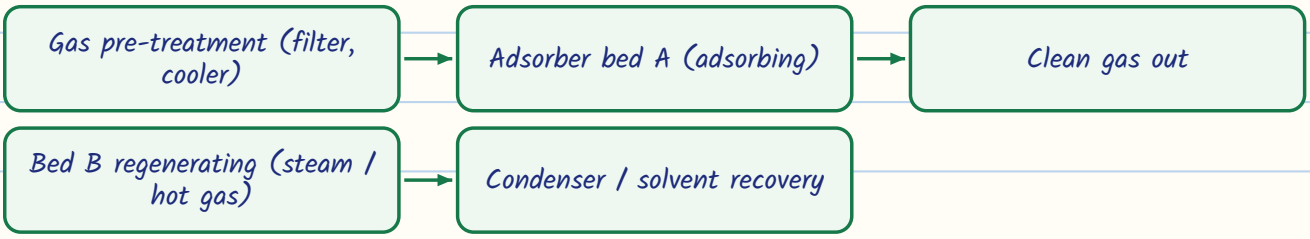
**Good adsorbent:** large surface area / porosity, high capacity and selectivity, mechanical strength, regenerable, cheap, low pressure drop, no catalytic activity (carbon can ignite with ketones).

## 4.2 Adsorption isotherms

- Show the amount adsorbed per unit mass of adsorbent (q or x/m) vs equilibrium concentration (or pressure) at constant temperature
- **Freundlich:** $q = K C^{1/n}$ (empirical; log q vs log C is a straight line)
- **Langmuir:** $q = q_m b C / (1 + b C)$ (monolayer; q -> $q_m$ at high C)
- **BET:** multilayer; used to determine the surface area
- Adsorption capacity increases with **higher concentration, higher pressure, lower temperature, larger surface area**; humidity reduces carbon capacity (keep RH < 50%)

## 4.3 Adsorption equipment

- **Fixed-bed adsorber (most common):** two or more beds - one adsorbs while the other is regenerated (**swing operation**); vertical or horizontal; gas velocity 0.2 - 0.5 m/s; bed depth 0.3 - 1.5 m; 1 - 3 kPa pressure drop
- **Moving-bed / fluidised-bed:** continuous flow of adsorbent, regenerated in a separate vessel; for high flow rate
- **Rotary wheel concentrator:** zeolite wheel adsorbs dilute VOC and desorbs in a small hot stream to an oxidiser
- **Canister / disposable cartridge:** small flows; deodorisation, gas masks



## 4.4 Breakthrough curve and bed design

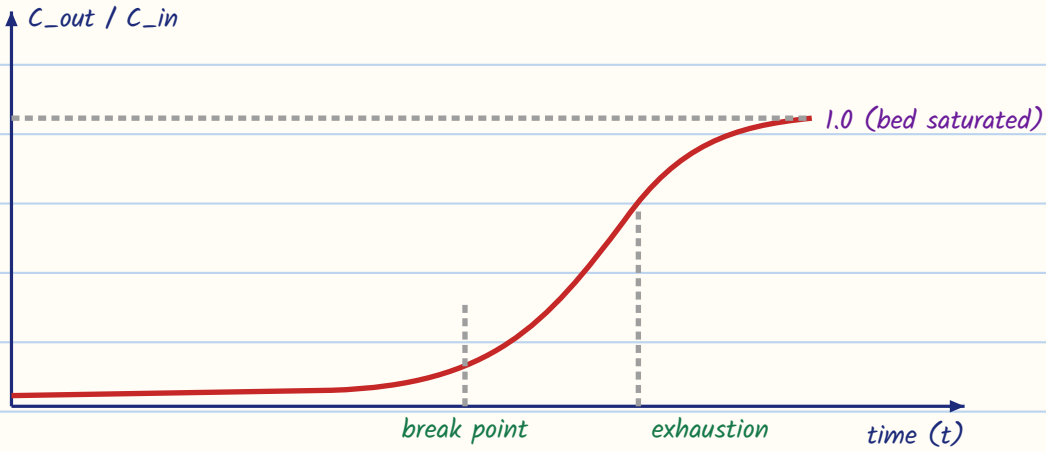



Fig. Breakthrough curve of a fixed bed

- Gas enters the bed; the **mass transfer zone (MTZ)** moves through the bed. When its front reaches the bed end, the outlet conc. begins to rise (**break point**, normally $C/C_0 = 0.05$). At **exhaustion** $C = C_0$ (bed saturated)
- A narrow MTZ means efficient use of the bed. Bed **working capacity** = 30 - 70% of the equilibrium capacity
- **Bohart-Adams / BDST equation:** $t_b = (N_0 / C_0 v) Z - (1 / K C_0) \ln (C_0 / C_b - 1)$ where $N_0$ = adsorption capacity per volume of bed, v = gas velocity, Z = bed depth, K = rate constant
- **Critical bed depth** $Z_c = (v / K) \ln(C_0 / C_b - 1)$: the minimum Z for any breakthrough time (t = 0)
- **Simple design:** mass of adsorbent = (mass of pollutant to be adsorbed per cycle) / (working capacity); e.g. 100 kg/h solvent x 8 h = 800 kg; with capacity 0.15 kg/kg -> **5300 kg** of carbon (per bed)

### 4.5 Regeneration

- **Thermal swing:** low-pressure steam (100 - 150 deg C) or hot inert gas drives off adsorbate; steam is condensed and solvent recovered by decanting / distillation; then bed is dried and cooled
- **Pressure swing / vacuum swing (PSA):** reduce pressure to desorb; used for small volume streams
- Chemical regeneration or replacement for chemisorbed gases; carbon lost by attrition, deactivated by high-boiling compounds
- **Advantages:** recovery of valuable solvent, 95 - 99% removal even for low concentrations, handles odours and mixtures, no wastewater (if dry)
- **Disadvantages:** pre-treatment needed (dust, moisture, T < 40 deg C), high cost of adsorbent, fire hazard in carbon beds (hot spots), disposal of spent adsorbent, high energy for regeneration

## 5. Combustion (Thermal and Catalytic Oxidation)

Organic vapours, CO and odorous gases are **oxidised (burnt)** at high temperature into $CO_2$ and $H_2O$: **$C_xH_y + (x + y/4)\ O_2 \rightarrow x\ CO_2 + (y/2)\ H_2O$ + heat**. Efficiency 95 - 99.99%. Needs good mixing.

**The three T's of combustion: TEMPERATURE (650-1100 deg C) + TIME (residence 0.3 - 1 s) + TURBULENCE (good mixing)**

| Equipment | Operation | Remarks |
|---|---|---|
| Direct flame (thermal) incinerator / afterburner | Gas heated to 650 - 1100 deg C with burner; residence 0.5 s | 95 - 99%; fuel cost high; halogens give HCl |
| Recuperative thermal oxidiser | Hot exit gas preheats feed through a heat exchanger (up to 70% recovery) | Lower fuel; fixed cost higher |

| Equipment | Operation | Remarks |
|---|---|---|
| Regenerative thermal oxidiser (RTO) | Ceramic beds store heat; flow switches alternately; 95% heat recovery | Low fuel for dilute VOC; 99% destruction |
| Catalytic oxidiser | Pt / Pd or metal oxide catalyst; 250 - 500 deg C; lower temperature and cost | Poisoned by S, Pb, P, Hg, Si; 90 - 98% |
| Flare | Open burning at an elevated or ground-level tip, steam-assisted to avoid smoke | Emergency, waste gas of refineries; 98% destruction; noise, light |
| Boilers / process heaters | Waste gas used as fuel supplement | Energy recovery with high residence time |

## Design aspects

- **Residence time:** $V = Q \times t$ ($Q$ at operating temperature; $t = 0.3 - 1$ s); length : dia ratio about 2
- **Heat balance:** $Q_{fuel} = m\ C_p\ (T_{out} - T_{in})$ (1 - heat recovery) - waste gas heating value can cover the load if VOC conc. is high
- **Safety:** keep VOC concentration below **25% of the lower explosive limit (LEL)** (use dilution air); flame arrestors, explosion vents
- **Heating value rule:** a stream with > 1.9 $MJ/m^3$ (50 Btu/scf) can sustain combustion without added fuel
- Problems: $NO_x$ formation (thermal $NO_x$ at > 1200 deg C), CO from incomplete burning, acid gases from Cl, S compounds -> scrubber needed after the incinerator

**Catalytic converter reaction:** $CO + 1/2\ O_2 \rightarrow CO_2$; $C_xH_y + O_2 \rightarrow CO_2 + H_2O$. Catalyst forms: honeycomb ceramic monolith or pellets; space velocity 10,000 - 50,000 per hour; life 3 - 5 years.

## 6. Condensation

**Condensation** removes a vapour from a gas stream by **lowering the temperature** (or raising the pressure) until the **partial pressure of the vapour exceeds its saturation vapour pressure**; vapour changes to liquid. It is best for **high concentration (> 5000 ppm) and high boiling** vapours (solvent recovery); often used as pre-treatment before adsorption, absorption or incineration.

### Types of condensers

| Type | Description | Remarks |
|---|---|---|
| Surface condenser (shell and tube / plate) | Coolant (water, brine) flows separately across tubes; condensate collected and does not mix | Product recovered pure; large area; common |
| Contact (direct) condenser | Coolant (water spray) mixes directly with vapour (spray tower, jet, barometric) | Cheap, compact; condensate contaminated with water |
| Refrigerated / cryogenic condenser | Chilled coolant (-10 to -40 deg C) or liquid $N_2$ (-196 deg C) | > 90 - 99% recovery of volatile VOC; costly |

## Design equations

- **Condensation ends when partial pressure of vapour = vapour pressure $p^*$ at condenser outlet temperature:** mole ratio $Y_{out} = p^* / (P - p^*)$
- **Recovery (efficiency): eta = $1 - Y_{out} / Y_{in}$**; $Y_{in} = y_{in} / (1 - y_{in})$
- **Heat duty:** $Q = m_g C_{pg} (T_1 - T_2) + m_v [ C_{pv} (T_1 - T_{dew}) + lambda + C_{pl} (T_{dew} - T_2) ]$
- **Heat transfer area:** $A = Q / (U \times DT_{lm})$ with $U = 100 - 500$ $W/m^2.K$ for surface condensers; coolant flow from heat balance
- Antoine equation for vapour pressure: $\log p^* = A - B / (T + C)$

**Example:** a stream has 10% (v/v) solvent vapour at 1 atm (760 mm Hg) and is cooled to a temperature at which $p^* = 20$ mm Hg. $Y_{in} = 0.10/0.90 = 0.111$; $Y_{out} = 20 / (760 - 20) = 0.0270$; **recovery = $1 - 0.0270/0.111 =$ 75.7%**. To get 90% recovery the $p^*$ must be only 8.5 mm Hg - a lower temperature.

- **Advantages:** solvent can be recovered and reused, no secondary waste in surface type, simple. **Limitations:** inefficient for dilute streams, large cooling load; non-condensable gas still carries vapour (needs afterburner / carbon bed polishing); mist formation

## 7. Comparison and Selection of Equipment

| Method | Gas conc. range | Efficiency | Recovery |
|---|---|---|---|
| Absorption | 250 - 10,000 ppm | 90 - 99% | Yes (by regeneration) |
| Adsorption | 50 - 10,000 ppm | 90 - 99% | Yes (solvent) |
| Thermal oxidation | 100 - 2000 ppm (< 25% LEL) | 95 - 99.99% | No (heat only) |
| Catalytic oxidation | 100 - 2000 ppm | 90 - 98% | No |
| Condensation | > 5000 ppm | 50 - 90% (99% cryogenic) | Yes (liquid product) |
| Biofilter | < 1000 ppm | 80 - 95% | No |

## 8. Quick Revision / Important Questions

1. Explain the methods of controlling $SO_2$ emission (FGD). Differentiate regenerative and non-regenerative processes.
2. Explain SCR and SNCR for $NO_x$ control.
3. Describe the principle and design of a packed absorption tower. What is HTU / NTU?
4. Compare physical and chemical adsorption. Explain adsorption isotherms.
5. With a neat sketch explain a fixed-bed adsorber; breakthrough curve and regeneration.
6. Explain thermal and catalytic incinerators and flares.
7. Explain the working of surface and contact condensers; give the design equations.

# AIR POLLUTION AND CONTROL

## Subject Code : 23A01504b

## UNIT - V : AUTOMOBILE AND INDOOR POLLUTION

**Syllabus:** Vehicular pollution - Sources and types of emission - Effect of operating conditions - Alternate fuels and emissions - Emission controls and standards, Strategies to control automobile pollution - Causes of indoor air pollution - Changes in indoor air quality - Control and air cleaning systems - Indoor air quality.

## PART A : AUTOMOBILE (VEHICULAR) POLLUTION

### I. Vehicular Pollution - Introduction

Road vehicles are **mobile, ground-level, line sources** that release pollutants close to people (breathing height), so their effect is much greater than the quantity suggests. In Indian metros vehicles contribute **30 - 70%** of CO, $NO_x$, HC and a major part of PM. Growth in vehicle numbers (> 30 crore registered), old vehicles, poor fuel quality, congestion and poor maintenance aggravate the problem.

- Fuel: petrol (spark ignition, SI engine, Otto cycle) and diesel (compression ignition, CI engine); two-stroke and four-stroke engines
- Complete combustion: $C_xH_y + (x + y/4)\ O_2 \rightarrow x\ CO_2 + (y/2)\ H_2O$
- Real engines give incomplete combustion: CO, HC, soot, $NO_x$, aldehydes in addition

# 2. Sources and Types of Emission

## 2.1 Sources of emission from a petrol vehicle

| Source | Share of HC | Pollutants |
|---|---|---|
| Exhaust pipe (tailpipe) | about 60% | CO, HC, $NO_x$, aldehydes, Pb, particulates, $SO_2$ |
| Crankcase blow-by | about 20% | Unburnt HC leaking past piston rings |
| Evaporative (fuel tank, carburettor) | about 20% | HC vapours (diurnal + hot soak + running losses) |
| Tyre, brake wear, road dust | - | PM (non-exhaust), heavy metals |

## 2.2 Pollutants and how they form

| Pollutant | Formation | Effects |
|---|---|---|
| Carbon monoxide (CO) | Incomplete combustion, rich mixture, low $O_2$; idling | Binds Hb; headache, death |
| Hydrocarbons (HC) | Unburnt fuel; flame quenching at cold walls; crevice volume; misfire | Smog, cancer (benzene, PAH) |
| $NO_x$ (NO, $NO_2$) | Thermal (Zeldovich) at > 1800 K, excess $O_2$: $N_2 + O \rightarrow NO + N$; $N + O_2 \rightarrow NO + O$ | Acid rain, smog, lung damage |
| Particulates (soot) | Diesel: fuel-rich pockets; cracking of fuel; also lube oil | Lung cancer; black smoke |
| Lead (Pb) | Tetra-ethyl lead anti-knock in petrol (phased out in India 2000) | Neurotoxic |
| $SO_2$ | Sulphur in fuel (diesel > petrol) | Respiratory, acid rain |
| $CO_2$ | Complete combustion; ~ 2.3 kg per litre petrol | Global warming |
| Aldehydes, PAH, benzene | Partial combustion, alcohol fuels | Irritant, carcinogenic |

### 2.3 Petrol versus diesel engine emissions

| Pollutant | Petrol (SI) | Diesel (CI) |
|---|---|---|
| CO | High (1 - 10%) | Low (< 0.5%) - lean mixture |
| HC | Medium - high | Low (but smell, aldehydes) |
| $NO_x$ | Medium | High (high compression ratio, excess air) |
| PM / smoke | Very low | **High** - black smoke |
| $SO_2$ | Low | Higher |
| $CO_2$ | Higher per km | Lower (better efficiency) |

**Two-stroke engines** (two wheelers, autos) emit 3 - 5 times more HC and PM than four-stroke because fresh charge escapes with exhaust (scavenging loss) and lubricating oil is burnt with the fuel (white smoke).

## 3. Effect of Operating Conditions

Emissions change with engine load, speed and especially **air-fuel (A/F) ratio**.

Stoichiometric A/F for petrol is about **14.7 : 1** by mass.

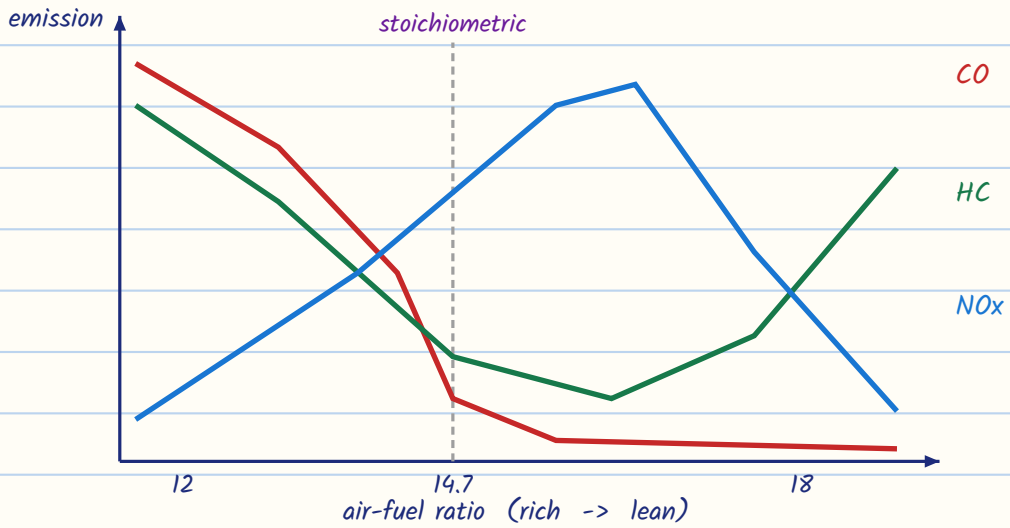



Fig. Variation of CO, HC and $NO_x$ emissions with air-fuel ratio (SI engine)

| Mode | Mixture / condition | CO | HC | $NO_x$ |
|---|---|---|---|---|
| Idling | Rich, low temperature, poor mixing | Very high | High | Low |
| Acceleration | Rich, then high temperature and load | High | Medium | High |

| Mode | Mixture / condition | CO | HC | $NO_x$ |
|---|---|---|---|---|
| Cruising (steady speed) | Lean, stable combustion | Low | Low | Medium - High |
| Deceleration | Throttle closed, high manifold vacuum; fuel film evaporates, misfire | Medium | Very high | Low |
| Cold start | Choke on; rich; catalyst not yet hot | Very high | Very high | Low |

- **Rich mixture (A/F < 14.7):** lack of $O_2$; CO and HC increase sharply, $NO_x$ is low (lower temperature)
- **Slightly lean (about 16 - 17):** maximum flame temperature -> **$NO_x$ peak**; CO and HC are minimum
- **Very lean (> 18):** misfire, unstable combustion -> HC rise; $NO_x$ falls (low temperature)
- **Ignition timing:** more advance -> more $NO_x$, less HC; retard reduces $NO_x$ but increases fuel consumption
- **Compression ratio, engine temperature, load and speed:** all increase $NO_x$; high speed reduces HC and CO per km but raises $NO_x$
- **Engine maintenance, age, tuning, altitude, fuel quality** - worn rings, dirty air filter, leaking injectors increase CO and HC
- **Traffic congestion** (stop-go) lengthens idling and acceleration modes; emission per km at 20 km/h is 2 - 3 times that at 60 km/h
- **Diesel engines:** smoke increases with load, as A/F decreases at full load; $NO_x$ increases with temperature and injection advance

## 4. Alternate Fuels and Emissions

Used to reduce emissions, dependence on crude oil and carbon footprint.

| Fuel | Features | Emission effects |
|---|---|---|
| **CNG** (compressed natural gas, mainly $CH_4$, stored at 200 bar) | High octane (130), mixes well; cheaper; used in Delhi buses, autos | CO -90%, HC (non-methane) -, $NO_x$ -, **no PM, no Pb, no $SO_2$**; $CO_2$ lower by 25%; methane is a GHG |
| **LPG / Autogas** (propane, butane) | Easy to liquefy at 8 bar; octane ~ 105 | Lower CO, HC, $NO_x$ than petrol; no PM |
| **Ethanol** (blends E10, E20 in India; bio-ethanol from sugarcane, grain) | Renewable, high octane; lower energy content; water-absorbing | Lower CO, HC; higher aldehyde (acetaldehyde); lower $CO_2$ life-cycle |
| **Methanol** (M85, M100) | From natural gas or coal / biomass; toxic; corrosive | Low $NO_x$, PM; formaldehyde emissions high |
| **Biodiesel** (FAME from jatropha, pongamia, used oil; BS - B20) | Renewable, biodegradable, sulphur-free, high cetane | Lower PM, CO, HC, $SO_2$; slightly **higher $NO_x$** |
| **Hydrogen** (IC engine / fuel cell) | Highest energy per kg; storage and cost problems | Only water in fuel cell; $NO_x$ in IC engine; zero CO, HC, $CO_2$ |
| **Electric vehicles (EV)** battery / hybrid / plug-in | Zero tailpipe emission; power from grid; FAME scheme | No local pollution; emissions shift to power plants unless renewable energy; battery disposal |
| Dimethyl ether (DME), producer gas, biogas, synthetic fuels | Diesel substitutes / blends | Low soot; limited use |

**CNG and EV are best for cities. BS-VI petrol and diesel (sulphur 10 ppm) enable after-treatment devices to work.**

# 5. Emission Controls

## 5.1 Engine modifications (in-engine controls)

- **Better combustion chamber design** (hemispherical, swirl, squish), higher turbulence; **lean burn** engine; **stratified charge**
- **Electronic fuel injection** (MPFI, GDI, common-rail diesel) with ECU and **lambda (oxygen) sensor** closed-loop control instead of carburettor
- **Exhaust Gas Recirculation (EGR):** 5 - 15% exhaust gas is returned to the intake; inert gas lowers peak flame temperature -> **$NO_x$ reduced by 50 - 70%**
- **Positive Crankcase Ventilation (PCV) valve:** blow-by gases returned to intake manifold -> crankcase HC removed
- **Evaporative emission control:** vapour from tank and carburettor is stored in an **activated carbon canister**; purged to intake when the engine runs
- **Turbocharging, intercooling, variable valve timing**, injection timing retard; high-pressure injection reduces smoke in diesel
- **Two-stroke:** direct injection (fuel injected after port closure) or conversion to four-stroke

## 5.2 Fuel modifications

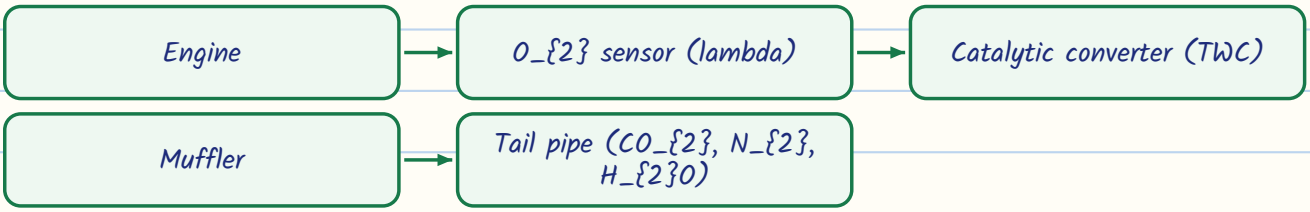
- Unleaded petrol (lead removed - allows catalyst); low-sulphur fuels (BS-VI: 10 ppm); low benzene (< 1%) and low aromatics; oxygenates (ethanol, MTBE) in blends; reformulated fuel; higher cetane diesel; fuel additives; clean alternative fuels

## 5.3 Exhaust after-treatment devices

**Three-Way Catalytic Converter (TWC):** a stainless steel shell with a ceramic honeycomb monolith (or metallic), coated with **alumina washcoat** impregnated with noble metals - **Pt, Pd (oxidation) and Rh (reduction)**. Operates near **stoichiometric** A/F (lambda = 1), temperature 300 - 800 deg C. Converts three pollutants:

- Oxidation of CO: $2\ CO + O_2 \rightarrow 2\ CO_2$
- Oxidation of HC: $C_xH_y + (x + y/4)\ O_2 \rightarrow x\ CO_2 + (y/2)\ H_2O$

- Reduction of $NO_x$: $2\ NO + 2\ CO \rightarrow N_2 + 2\ CO_2$; $2\ NO + 2\ H_2 \rightarrow N_2 + 2\ H_2O$
- Efficiency > 90% when hot; **poisoned by lead, sulphur, phosphorus** (so unleaded fuel needed); needs $O_2$ sensor




- **Diesel Oxidation Catalyst (DOC):** oxidises CO, HC and soluble organic fraction of PM
- **Diesel Particulate Filter (DPF):** wall-flow ceramic filter traps > 95% of soot; regenerated by burning soot at 550 - 600 deg C (active / passive)
- **SCR with urea (AdBlue / DEF):** 32.5% urea solution injected into the exhaust; $NH_3$ reduces $NO_x$ to $N_2$ on catalyst (heavy diesel vehicles; BS-VI)
- **Lean $NO_x$ trap, thermal reactor / secondary air injection** (older engines), **OBD (On-Board Diagnostics)** monitors the control system
- **Inspection and maintenance (I/M):** PUC (Pollution Under Control) certificate - CO and HC test (petrol), smoke opacity test (diesel, Hartridge units)

## 6. Emission Standards

Controlled in India under the **Motor Vehicles Act 1988, Central Motor Vehicles Rules 1989**, Air Act 1981 and Environment (Protection) Act 1986. **Bharat Stage (BS) norms** (introduced 2000) are based on European (Euro) norms.

| Norm | Equivalent | Year (nationwide) |
|---|---|---|
| India 2000 / BS-I | Euro 1 | 2000 (all India) |
| BS-II | Euro 2 | 2001 (NCR), 2005 nationwide |
| BS-III | Euro 3 | 2005 (cities), 2010 nationwide |
| BS-IV | Euro 4 | 2010 (13 cities), 1 April 2017 nationwide |
| BS-V | Skipped | Leap-frogged to BS-VI |
| **BS-VI** | Euro 6 | **1 April 2020** nationwide (phase-2 RDE from April 2023) |

| Passenger car limit (g/km) | BS-IV | BS-VI |
|---|---|---|
| Petrol: CO | 1.0 | 1.0 |
| Petrol: $NO_x$ | 0.08 | 0.06 |
| Diesel: $NO_x$ | 0.25 | 0.08 |
| Diesel: PM | 0.025 | 0.0045 |
| Fuel sulphur | 50 ppm | 10 ppm |

- BS-VI introduced **DPF, SCR, OBD, GDI particle limits** (PM and particle number), real-driving emission test and evaporative control
- **Other:** noise limits; CAFE (fuel economy) norms; age limit and **vehicle scrappage policy**; mass emission standards for 2-3 wheelers; tests on chassis dynamometer using the Indian Driving Cycle / WLTP, with CVS sampling

## 7. Strategies to Control Automobile Pollution

| Strategy | Measures |
|---|---|
| Technological | Catalytic converters, EFI, EGR, DPF; BS-VI norms; hybrid and electric vehicles; retrofitting of old buses; good maintenance |
| Fuel | Unleaded, low-sulphur, CNG / LPG, ethanol blend, biodiesel, hydrogen; fuel adulteration check |
| Traffic management | Synchronised signals, one-way roads, flyovers, bypass / ring roads, no-parking zones, speed limits, congestion pricing, odd-even scheme |
| Public transport | Metro rail, BRT, e-buses, last-mile connectivity, cycling and pedestrian lanes, car pooling |
| Urban planning | Mixed land use, green belt, plantation along roads (pollution sink), buffer zones, tall buildings away from roads |
| Legal / economic | PUC checks, age limit for vehicles, scrappage incentive, higher tax on old and diesel vehicles, EV subsidy (FAME), parking fee |
| Awareness | Drive less, switch off engine at red light, tyre pressure, correct driving |

# PART B : INDOOR AIR POLLUTION

## 8. Introduction to Indoor Air Quality

People spend **80 - 90% of their time indoors** (home, office, school, vehicles). **Indoor air pollution** is the presence of contaminants in the air inside buildings that may harm the health or comfort of occupants. Indoor levels of many pollutants are **2 - 5 times (sometimes 100 times) higher** than outdoors (US-EPA). The WHO estimates that **household air pollution** (cooking with solid fuel) causes about 3 million premature deaths per year, mostly in developing countries (women and children).

- **Indoor Air Quality (IAQ)** = the nature of the air that affects health and well-being of occupants
- **Sick Building Syndrome (SBS):** > 20% of occupants complain of headache, eye / nose / throat irritation, fatigue, dizziness, nausea - symptoms disappear on leaving the building, cause not specific
- **Building Related Illness (BRI):** a specific diagnosable disease from the building: Legionnaires' disease, hypersensitivity pneumonitis, asthma, humidifier fever

## 9. Causes of Indoor Air Pollution

| Source | Pollutants | Health effects |
|---|---|---|
| **Combustion** - chulha, wood, dung, coal, kerosene, gas stove, heaters | CO, PM2.5, $NO_2$, $SO_2$, PAH, formaldehyde | COPD, lung cancer, ALRI in children, eye disease |
| **Tobacco smoke (ETS)** | PM, CO, nicotine, benzene, > 60 carcinogens | Cancer, asthma, heart disease |
| **Building materials**: asbestos, pressed wood, insulation, paint, adhesives | Asbestos fibres, formaldehyde, VOCs, Pb | Asbestosis, mesothelioma, irritation, cancer |
| **Furnishing**: carpets, foam, furniture, cleaning products, air-fresheners | VOCs (toluene, xylene), phthalates, ozone | Headache, allergic reaction |
| **Soil / building stone** | **Radon** (Rn-222, from decay of uranium - radium) | Lung cancer (2nd cause after smoking) |
| **Biological**: damp walls, AC coils, humidifiers, pets, carpets | Mould spores, dust mites, bacteria, Legionella, pollen, dander | Allergy, asthma, infections |
| **Pesticides, mosquito coils, incense, candles** | PM, VOC, pesticides | Neurological, respiratory |
| **Office equipment**: printers, photocopiers | Ozone, VOC, toner particles | Irritation |
| **Occupants** (breathing, perspiration) | $CO_2$, odours, bacteria, viruses | Stuffiness, infection spread |
| **Infiltration of outdoor air** | Traffic fumes, dust, pollen, industrial gases | Many |

- **Poor ventilation** is the key factor: pollutants accumulate. Also high temperature and humidity, poor maintenance of HVAC and moisture problems (leaks, condensation)

## 10. Changes in Indoor Air Quality

- **Energy-efficient (tight) buildings** since 1970s oil crisis: sealed windows, low air-exchange rate (0.2 - 0.5 ACH) -> pollutants build up
- **Increase in synthetic materials** (plastics, composites, solvents, cleaners, electronics) -> more VOCs
- **Time of day / season:** cooking peaks, winter heating and closed windows give higher concentration; humidity in monsoon promotes mould
- **Urbanisation and outdoor pollution:** infiltration of PM2.5; **I/O ratio** (indoor / outdoor) shows if the source is indoor (> 1) or outdoor (< 1)
- **Changes in lifestyle:** more time indoors, AC (recirculation), air fresheners, new furniture ('new building smell')
- **Occupancy and activity:** $CO_2$ rises with number of people; $CO_2$ > 1000 ppm indicates inadequate ventilation (outdoor 400 ppm)
- **Chemical reactions indoors:** ozone + terpenes (cleaning fluids) -> secondary organic aerosol, formaldehyde

### Mass balance model for a room (well-mixed)

**$V\,dC/dt = Q\,C_0 - Q\,C + S - k\,V\,C$ -> steady state: $C = (Q\,C_0 + S) / (Q + k\,V)$**

- $V$ = room volume ($m^3$), $Q$ = ventilation flow ($m^3/h$) = ACH x $V$, $C_0$ = outdoor conc., $S$ = source emission rate (mg/h), $k$ = removal / decay rate (1/h)
- **Example:** $V$ = 50 $m^3$, ACH = 0.5 -> $Q$ = 25 $m^3/h$; source $S$ = 5 mg/h; $C_0$ = 0, $k$ = 0 -> **$C = S / Q$ = 0.2 $mg/m^3$**. Doubling ACH halves the concentration.

| Pollutant | Guideline / limit (typical) |
|---|---|
| $CO_2$ | 1000 ppm (comfort indicator) |
| CO | 9 ppm (8 hr); WHO 7 $mg/m^3$ (24 hr) |
| Formaldehyde | 0.1 $mg/m^3$ (30 min, WHO) |
| Radon | 100 $Bq/m^3$ reference level (WHO); action at 300 |
| PM2.5 / PM10 | WHO 24 hr: 15 / 45 $ug/m^3$ |
| Temperature, RH | 23 - 26 deg C, 30 - 60% (comfort) |

**Ventilation standards:** ASHRAE 62.1 - about **7.5 L/s per person** (15 cfm) in offices; **National Building Code of India (NBC)**, IS 3362 (natural ventilation); IGBC / Green building guidelines for IAQ.

## II. Control of Indoor Air Pollution and Air Cleaning Systems

### II.1 Control strategies (in order of preference)

1. **Source control** (best): eliminate or reduce the source - no smoking indoors, **clean cooking fuel (LPG, biogas, electric, improved chulha with chimney)** - PM Ujjwala Yojana, low-VOC paints, formaldehyde-free boards, asbestos removal, store solvents outside, proper pest control
2. **Ventilation:** dilute the pollutants by outdoor air - natural (windows, vents, cross-ventilation), mechanical (exhaust fans, kitchen chimney / hood, bath exhaust), **heat recovery ventilator (HRV / ERV)**; ACH 0.5 - 1.0 for homes; offices 4 - 6 ACH; provide a 'flush-out' before occupying a new building
3. **Air cleaning:** remove pollutants from the air by equipment (when source control and ventilation are not enough)
4. **Control of humidity** (30 - 60%), fix leaks, cleaning, HVAC maintenance, filter change, **radon mitigation** (sealing cracks, sub-slab depressurisation fan), education of occupants

## 11.2 Air cleaning systems

| Device | Principle | Removes | Remarks |
|---|---|---|---|
| **Mechanical filters** (flat, pleated, bag; MERV rating) | Impaction, interception, diffusion | Dust, pollen, mould | Cheap; clog; dP |
| **HEPA filter** | Fibre mat; 99.97% for 0.3 um | PM2.5, allergens, bacteria | High dP, costly; most used in homes, hospitals, clean rooms |
| **Electronic air cleaner (ESP type)** | Charge + collecting plates | Fine PM, smoke | Low dP; may emit ozone; wash plates |
| **Activated carbon filter** | Adsorption | VOC, odours, formaldehyde, ozone | Saturates; replace / regenerate |
| **UVGI** (UV-C light) | Kills micro-organisms by DNA damage | Bacteria, viruses, mould | Coil and duct disinfection; needs exposure time |
| **Photocatalytic oxidation ($TiO_2$ + UV)** | Oxidises VOC to $CO_2$ + $H_2O$ | VOC, odour | By-products possible |
| **Ionisers / ozone generators** | Ions / ozone | Particles (ionisers) | **Not recommended** - $O_3$ is harmful |
| **Botanical (plants)**, biofilters | Phytoremediation | Some VOC | Very small effect |
| **Chemical / gas-phase** ($KMnO_4$ on alumina) | Chemisorption | Acid gas, formaldehyde | Industrial |

- **Clean air delivery rate (CADR)** = flow rate x efficiency ($m^3/h$) indicates cleaner capacity; choose CADR ~ 2/3 of the room area
- Portable cleaners work on re-circulated air: the effective removal rate $k_c$ = CADR / V adds to the model above (C reduces by factor (Q + kV))

- Filters need regular maintenance; poor maintenance makes the filter a pollution source

## 12. Indoor Air Quality (IAQ) Management

- **Assessment:** walk-through inspection, occupant questionnaires, measurement of temperature, RH, $CO_2$, CO, PM2.5, TVOC, HCHO, radon, microbes (sampling with impactor, passive samplers, detectors)
- **IAQ standards:** WHO guidelines for indoor air quality; ASHRAE 62.1 & 55; Indian Green Building Council (IGBC) and the Bureau of Indian Standards (BIS) norms; Smoke-free policy
- **Management plan:** design (material selection, HVAC sizing, ventilation), commissioning, operation and maintenance, monitoring and complaint handling, communication
- Benefits: improved health, productivity and comfort; lower absenteeism, energy-saving through demand-controlled ventilation ($CO_2$ sensors)

## 13. Quick Revision / Important Questions

1. Explain the sources and types of emission from a petrol vehicle. Compare petrol and diesel pollutants.
2. Discuss the effect of operating conditions (A/F ratio, speed, load) on emission.
3. Write about alternate fuels (CNG, ethanol, biodiesel, hydrogen, EV) and their emissions.
4. Explain the three-way catalytic converter with a sketch. What are EGR and PCV?
5. What are Bharat Stage norms? List the strategies to control automobile pollution.
6. What are the causes of indoor air pollution? Explain SBS and BRI.
7. Explain the control methods and air-cleaning systems of indoor air quality (HEPA, carbon, UVGI).